



C.H.E.C.K: COMMUNITY HEALTH EVALUATION CENTER KIOSK

RUBANG, VINCE MARC JUSTINE O.

HERMOSURA, DHON MARCK V.

LANGOEY, WINFREY CLINT O.

MELWYN MARC O. RAMOS

PASIBE, ASHLEY A.

ILOCOS SUR POLYTECHNIC STATE COLLEGE
CANDON CAMPUS
DARAPIDAP, CITY OF CANDON, ILOCOS SUR

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

NOVEMBER 2025



APPROVAL SHEET

This Capstone Project entitled “C.H.E.C.K Community Health Evaluation Center Kiosk” was prepared and submitted by **VINCE MARC JUSTINE O. RUBANG, WINFREY CLINT O. LANGOEY, DHONMARCK V. HERMOSURA, MELWYN MARC O. RAMOS, ASHLEY A. PASIBE** in partial fulfillment of the requirement for the degree, Bachelor of Science in Information Technology has been examined by the ORAL EXAMINATION COMMITTEE (OREC) on November, 2025 composed of;

SONNY ROBERT A. LITUSQUEN

Member

CLARIZA G. VALDEZ, MIT

Member

ZJ G. VILLANUEVA, MSE

Statistician Expert

RALPH NICOLE A. SEVILLA, MSE

English Critic

FE AIRAM R. DAKIG, MIT

Capstone Adviser

ARIEL A. DELA CRUZ, MIT

Chairman

Accepted and approved in partial fulfillment of the requirements for the degree,
Bachelor of Science in Information Technology.

ARIEL A. DELA CRUZ, MIT

Program Head

ANNIE D. DORADA, Ed.D

Campus Director

ABSTRACT

VINCE MARC JUSTINE O. RUBANG, WINFREY CLINT O. LANGOEY, DHONMARCK V. HERMOSURA, MELWYN MARC O. RAMOS, ASHLEY A. PASIBE., “C.H.E.C.K “Community Health Evaluation Center Kiosk” (C.H.E.C.K: IoT Health Check Kiosk for Community Vital Monitoring). Ilocos Sur Polytechnic State College, Darapidap Candon City, Ilocos Sur. December 2025.

Adviser: **DAKIG, FE AIRAM R., MIT**

Barangay Darapidap, a community in Candon City, Ilocos Sur, relies on traditional manual methods for checking vital signs such as height, weight, BMI, heart rate, and temperature, a process that is time-consuming and prone to human error. These manual procedures often result in delayed access to medical records and hinder timely intervention. To address these challenges, C.H.E.C.K aims to deliver a localized, technology-driven solution by automating vital sign monitoring and streamlining data management. The project seeks to reduce the workload of healthcare workers, improve the accuracy and speed of health assessments, and promote preventive care in local communities. The researchers employed a developmental research design to create and refine the system. The Agile Development Methodology, specifically the Scrum framework, was adopted to guide the development process, segmenting the work into five core phases: Initiation, Planning and Estimation, Implementation, Review and Retrospective, and Release. This iterative approach ensured adaptability and continuous improvement based on user feedback. To evaluate the system's acceptability, an evaluation sheet based on the ISO 9126-2 standard was used, measuring six indicators: Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability. In conclusion, the researchers successfully developed C.H.E.C.K, an IoT Health Check Kiosk. The system integrates sensors for temperature, heart rate, oxygen level, height, and weight measurements into one device, utilizes biometric scanning for user authentication, and provides a real-time web dashboard for authorized health personnel. The evaluation of C.H.E.C.K showed high user acceptability, with a grand mean of 6.72, which corresponds to “Strongly Agree”. The system's greatest strengths were identified in Efficiency, Functionality, and Usability. The study provides a foundation for exploring health technologies that are efficient, affordable, and localized. The researchers recommend that future researchers focus on the integration of additional features to improve and adapt similar systems.

Keywords: ISPSC, Candon, IoT, Health Kiosk, Agile Scrum, ISO 9126-2, Vital Monitoring, Biometrics



ACKNOWLEDGEMENT

The researchers would like to express their sincere gratitude to all the individuals who contributed to the success of this study. Special recognition is extended to:

Dr. Annie D. Dorada, Campus Director of Ilocos Sur Polytechnic State College, for her unwavering support, dedication, and commitment to providing quality Education to ISPSCians;

Mr. Ariel A. Dela Cruz, the Program Head of Bachelor of Science in Information Technology, for his tireless guidance, valuable technical advice, and immense support throughout the planning, development, and completion of the C.H.E.C.K project;

Ms. Fe Airam R. Dakig, for her essential role as Capstone Adviser, providing crucial insights, constructive feedback, and motivation that were instrumental in shaping the methodology and final output of the C.H.E.C.K system;

Mr. Ralph Nicole A. Sevilla, the English Critic, for his meticulous review of the manuscript, ensuring its clarity, coherence, and professional presentation;

Mr. ZJ G. Villanueva, the Statistician, for his time, expertise, and guidance in providing the necessary statistical analysis and ensuring the reliability of the research findings;



Their **instructors**, for imparting the foundational knowledge and technical skills that were essential for the conceptualization and development of the IoT system;

Their **Parents and Families**, for their enduring love, moral support, and financial sacrifice, which served as the primary motivation for completing this academic endeavor;

Their **Friends and Classmates**, for the deep camaraderie, encouragement, shared challenges, and collaborative spirit that made the demanding process of research enjoyable and manageable;

Above all, to the **Almighty God**, for the strength, protection, wisdom, and countless blessings bestowed upon them throughout the duration of the study.

V.M.J.O.R.

W.C.O.L.

D.M.V.H.

M.M.O.R.

A.A.P.



DEDICATION

This project represents a personal redemption, a triumphant culmination of a journey that was postponed by a two-year delay in my academic path. As the saying goes, "Delay is not denial." Those years were not wasted; they taught me essential lessons in resilience, patience, and the true value of an education. Finishing this complex work, despite the intense pressures and setbacks, serves as a powerful statement that I can overcome any obstacle placed before me and ultimately determine the outcome of my own life.

The road to completion was marked by countless struggles, which included the relentless cycle of sleepless nights fueled by debugging challenges and excessive energy drinks. This outcome is a product of navigating technical ups and downs, transforming a challenging concept into a functional, real-world system. This accomplishment is the proof of my perseverance and the commitment I made to myself that I would see this through, no matter the cost, turning past frustration into current pride.

This achievement is also dedicated to my unshakeable pillars of support: my Family and Friends. Above all, to my incredible Mother, whose sacrifices, emotional fortitude, and unwavering financial and moral support ensured that I would be here today. She is the reason I never truly gave up; her tireless belief was the constant safety net that prevented me from falling. To her, I owe the very possibility of standing at this finish line.

VINCE MARC JUSTINE O. RUBANG



DEDICATION

This capstone project, the culmination of years of study and dedication, is the realization of a long-held academic ambition. It is with immense gratitude and deep respect that I dedicate this work. This project would not have been possible without the invaluable lessons learned and the unwavering belief system instilled in me by those closest to my heart.

This work is lovingly dedicated to my parents, Mr. Edwin Langoey and Mrs. Ela Langoey, whose unwavering support, sacrifices, and guidance have shaped me into who I am today. Thank you for believing in me, for providing strength during challenges, and for continuously encouraging me to pursue my dreams.

To my wife and daughter who stands as my foundation and inspiration, and to everyone who has inspired me on this journey my teachers who shared knowledge with patience, my friends who stayed with me through every deadline and challenge, and those who continuously motivate me to become better this achievement is also yours.

Above all, I give thanks to God for granting me the wisdom, health, and opportunities to reach this chapter of my life. With faith, perseverance, and gratitude, I dedicate this milestone to all who have guided and supported me along the way.

WINFREY CLINT O. LANGOEY



DEDICATION

With deepest gratitude, I dedicate this Capstone Project to my core support system. To my loving family, particularly my parents, Alice Joy Hermosura and Ricardo Hermosura, thank you for your enduring sacrifices, constant encouragement, and for being the unwavering foundation of my academic success. Your belief in me, even during moments of profound self-doubt, was the strength I needed to complete this journey.

My sincere appreciation is extended to those who provided comfort and collaboration. To my girlfriend, Faye Louisse Shante Somera, your patience, understanding, and motivation offered essential support during the most challenging days. To my friends in school, thank you for the shared effort, teamwork, and laughter that made every obstacle lighter. I am also grateful to my instructors, whose guidance and dedication shaped my knowledge and inspired me to strive for excellence.

Finally, this project stands as a testament to personal resilience. Above all, to Almighty God, thank You for the wisdom and strength provided. And to myself, Dhon Marck V. Hermosura thank you for choosing not to give up. This Capstone is the culmination of hard work, faith, and the continuous support received along the way.

DHONMARCK V. HERMOSURA



DEDICATION

This study is dedicated to our families, whose unwavering support, patience, and belief in us became our source of strength throughout this journey.

Their encouragement kept us going, even in moments when the challenges felt overwhelming. We also extend this dedication to our teachers and mentors, whose guidance and commitment have shaped not only this project but also who we are as learners.

Their wisdom and encouragement pushed us to grow, improve, and pursue excellence in the field of IT. Lastly, we dedicate this work to the community of Barangay Darapidap. Their needs inspired the purpose of this project, and we sincerely hope that this system will serve as a meaningful contribution to improving their health services and overall well-being.

MELWYN MARC O. RAMOS



DEDICATION

This capstone project, C.H.E.C.K Community Health Evaluation Center Kiosk, is dedicated to all the people who have continually inspired and strengthened me throughout this journey.

To my dear family, thank you for your endless love, support, and sacrifices. Your encouragement has been my foundation and has kept me going, even in difficult times. To my instructors and mentors, I am grateful for your guidance, patience, and valuable insights. Your efforts helped shape this project and pushed me to grow and aim higher. To my friends, thank you for your support, assistance, and belief in me. Your presence made this journey easier and more meaningful.

Above all, I give thanks to Almighty God for the strength, wisdom, and protection He has provided from the beginning until this work was completed. Lastly, I dedicate this to myself for staying strong, working hard, and never giving up. This project is a testament to all the love, support, and contributions that helped make it possible.

ASHLEY A. PASIBE



TABLE OF CONTENTS

PREMILINARIES	Page
Title Page.....	i
Approval Sheet.....	ii
Abstract.....	iii
Acknowledgement.....	iv
Dedication.....	v
Table of Contents.....	x
List of Appendices.....	xii
List of Tables.....	xi
List of Figures.....	xiv
 CHAPTER 1 INTRODUCTION	
Background of the Study.....	1
Conceptual Framework.....	8
Research Paradigm.....	9
Objectives of the Study.....	10
Scopes and Limitations of the Study.....	11
Importance of the Study.....	12
 CHAPTER 2 METHODOLOGY	
Research Design.....	13
Project Plan.....	16
Project Assignment.....	19
Research Instrument.....	22
Data Analysis.....	23



CHAPTER 3 RESULTS AND DISCUSSIONS

Findings.....	25
Conclusion.....	49
Recommendations.....	50
GLOSSARY.....	52
REFERENCES.....	55
APPENDICES.....	61
BIOGRAPHICAL SKETCH.....	82



LIST OF APPENDICES

Appendix	Page
A. Letters and Communication.....	62
B. Evaluation/Tools/Instruments.....	63
C. Use Case Diagram.....	64
D. Context Diagram.....	65
E. Data Flow Diagram.....	66
F. Entity Relationship Diagram.....	67
G. User Manual.....	68
H. Actual and Final Design of C.H.E.C.K Kiosk.....	75
I. Certificate of English Critic.....	76
J. Certificate of Statistician.....	77
K. Certificate of Implementation.....	78
L. Pictures taken during Data Gathering, Planning and Development....	79



LIST OF TABLES

Tables	Page
1. Role Requirements and Responsibility.....	20
2. Distribution of Respondents.....	22
3. Level of Agreement or Disagreement of Users Response.....	24
4. Functionality of the System.....	43
5. Reliability of the System.....	44
6. Usability of the System.....	45
7. Efficiency of the System.....	46
8. Maintainability of the System.....	47
9. Portability of the System.....	48
10. Summary of Evaluation of the Developed System.....	49



LIST OF FIGURES

Figures	Page
1. Research Paradigm.....	9
2. Scrum Lifecycle.....	14
3. Gantt Chart.....	19
4. Block Diagram of the CHECK System.....	26
5. Esp32 Microcontroller.....	28
6. Fingerprint Sensor.....	29
7. MLX90614 Temperature Sensor.....	29
8. MAX30102 Pulse Oximeter Sensor.....	30
9. VL52L0X Time of Flight Sensor.....	31
10. Load Cell (200kg) + Amplifier.....	31
11. TFT Lcd Display.....	32



LIST OF PLATES

Plates	Page
1. Landing Page.....	33
2. Admin Dashboard.....	34
3. Admin BHW Management.....	35
4. Admin Resident's Management.....	36
5. Admin Vital Monitoring.....	37
6. Admin Support Manual.....	38
7. BHW Dashboard.....	38
8. BHW Resident's Management.....	39
9. BHW Vital Monitoring.....	40
10. BHW Support/Manual.....	41



Chapter 1

INTRODUCTION

Background of the Study

The increasing global demand for healthcare services has made it extremely difficult to control patient flow and deliver prompt treatment in emergency situations. Traditional manual check-in and triage methods are slow and ineffective, and their continued usage is frequently associated with problems like congestion, lengthy wait times, lower patient satisfaction, and higher mortality rates. (Sehgal et al., 2024). A formative evaluation of eHealth systems in geographically isolated and disadvantaged areas (GIDAs) in the Philippines found that intermittent electricity and unstable internet connectivity severely disrupted system operations, forcing staff to revert to manual records and delaying data transmission and patient care (Evio et al., 2024). One of the main latest implications of the Philippines is insufficient medical equipment and facilities that lead to the well-being situation of the persons in provincial areas. Every single citizen must receive equal medical attention, but one of the factors that they did not is the access because of the distance (Tolentino et al., 2021).

Health kiosks offer a multifaceted solution to address healthcare challenges in remote and resource-poor areas where there are often poor infrastructure and a lack of qualified health personnel. These kiosks can provide interactive health education through user-friendly interfaces that inform populations about the risk factors, prevention, monitoring, and management of chronic diseases. In addition, health kiosks can play a critical role in capturing epidemiological data in rural and hard-to-reach communities, contributing valuable, evidence-based information that can support

national and regional health programs and policymaking in developing countries (Joshi et al., 2013)

Remote consultations, real-time data sharing, and advanced health analytics have been made possible by the merging of telemedicine and artificial intelligence, which replaced the need for recurring clinical check-ups in traditional health monitoring. But there are still issues with guaranteeing accurate, ongoing monitoring, safeguarding data security, and increasing access to underprivileged areas (Suleiman & Adinoyi, 2023). Monitoring a person's body temperature and heart rate is essential, as these vital signs can indicate several serious health issues, including abnormal blood pressure, flu, high cholesterol, and other potential complications (Saxena et al., 2023).

Regular health monitoring is essential for maintaining good health, but long hospital queues and the need for ambulatory monitoring in our fast-paced world underscore the necessity for a simple health monitoring system adaptable to various settings. This system addresses the increasing demand for monitoring primary health parameters, particularly among older individuals, where conventional healthcare infrastructure often falls short. To overcome these challenges, the researchers proposed a comprehensive health monitoring system leveraging Internet of Things (IoT) technology. By utilizing IoT, they can wirelessly capture and transmit health data, ensuring interoperability. (EDP Sciences, 2024).

The primary challenges facing rural healthcare include poverty, high transportation costs, and the long travel times required to reach distant medical facilities. These issues are further worsened by a low doctor-to-population ratio, particularly in rural areas, which increases the risk of exposure to infectious diseases and contributes to hospital overcrowding (Ng et al., 2021). Addressing these obstacles is essential to improving healthcare access and health outcomes for people living in

rural communities. With the help of IoT technology, this remote patient health monitoring system wirelessly records and transmits critical health parameters. This allows for real-time data analysis, storage, and automated alerts to medical professionals, which lessens the strain on healthcare facilities and enhances prompt intervention.

The emergence of new diseases with high rates of transmission and significant health hazards in recent years has highlighted the increasing demand for cutting-edge technologies that support healthcare services both locally and remotely. The creation of an Internet of Things-based system for tracking vital health indicators is one such remedy. Vital indicators like heart rate, blood pressure, oxygen saturation, and body temperature can be measured by this technology both locally and remotely. First, local sensors are used to collect the measurements for each parameter. A particle photon microcontroller receives these readings, processes them, and transforms them into precise measurement values. After processing, the data is instantly saved to the cloud, allowing for convenient remote access and real-time monitoring (Laureano, 2024).

Advances in built-in Internet technology are crucial for remote treatment because of the modern lifestyle, the rapid acceptance of fast food, and environmental changes that are causing chronic, life-threatening diseases. In a long-term diagnostic procedure, the proposed work discusses an Internet of Things (IoT) paradigm that includes Wireless Health Sensors (WHS) that enable to observe biometric parameters that are essentially continuous and/or should be monitored, such as blood pressure, body temperature, electrocardiogram activity, pulse rate, and pulmonary functional quality. This allows to control future severity through predictive analysis via smart healthcare systems (Aski et al., 2019).

Health kiosks are publicly accessible computing devices that provide access to services, including health information provision, clinical measurement collection, patient self-check-in, telemonitoring, and teleconsultation. Although the increase in internet access and ownership of smart personal devices could make kiosks redundant, recent reports have predicted that the market will continue to grow (Austin et al., n.d.). Medical kiosks are easily accessible computing devices that can provide health information, conduct clinical measurements, facilitate self-monitoring, and enable remote consultation. Wider adoption of self-kiosks could ease patient management, particularly in resource-constrained or high-demand healthcare environments (Sen et al., 2025).

According to Romanick-Schmiedl and Raghu (2020), telemedicine is the application of technology to assist in patient diagnosis, treatment, monitoring, and management. It combines more conventional forms of communication like text messages, emails, and video chats with more contemporary tools like mobile health apps, video teleconferencing, and remote patient monitoring (Noceda et al., 2023). Many people believe that telemedicine is a safe, effective, and economical substitute for in-person medical care that benefits both patients and healthcare professionals. It lessens the burden on health systems and conserves important resources by enabling patients to obtain high-quality treatment for non-urgent diseases without going to a medical facility (Noceda et al., 2023; Romanick-Schmiedl & Raghu, 2020). Additionally, recent research shows how well it works to enhance patient outcomes and reduce readmission and hospitalization risks (Noceda et al., 2023).

Vital signs are measured and sent to distant specialists using the RxBox, a telemedicine gadget. In order to reduce morbidity and mortality from common diseases and inadequate access to care, it has been implemented in primary care health clinics

(PCHC) that serve underprivileged populations in the Philippines. However, it is still unknown what factors influence healthcare personnel's acceptance of this technology (Acta Medica Philippina, 2022).

Body temperature is a fundamental parameter for assessing an individual's health status. Elevated or abnormally low temperatures can indicate infections, inflammation, or other underlying conditions. IoT-based systems often use non-invasive temperature sensors, such as those integrated into RFID-enabled devices, to ensure continuous and comfortable monitoring (Kumar et al., 2024) (Jonnalagadda et al., 2023) (Mishra et al., 2024).

Blood pressure is a critical indicator of cardiovascular health. High blood pressure (hypertension) is a leading risk factor for heart disease, stroke, and kidney disease. IoT health monitoring systems often incorporate blood pressure sensors that provide accurate and real-time readings, enabling early detection of abnormalities (Kumar et al., 2024) (Laureano, 2024) (Ghazaly & Jain, 2023).

Heart rate monitoring is essential for assessing cardiac health. Abnormal heart rates, such as tachycardia or bradycardia, can signal conditions like arrhythmias, heart failure, or electrolyte imbalances. IoT systems typically use pulse sensors to measure heart rate, often in conjunction with blood pressure monitoring (Kumar et al., 2024) (Laureano, 2024) (Saxena et al., 2023).

Oxygen saturation is a vital parameter for respiratory health. Low SpO₂ levels can indicate conditions such as asthma, chronic obstructive pulmonary disease (COPD), or COVID-19. IoT-enabled systems often include SpO₂ sensors to monitor oxygen levels in real-time, ensuring timely intervention for individuals with respiratory distress (Dey & Ghosh, 2023) (Mishra et al., 2024).

The integration of RFID technology with IoT-enabled health check kiosks offers several advantages, including non-invasive monitoring, real-time data transmission, and seamless connectivity to cloud platforms for data storage and analysis. These systems often utilize low-cost, miniaturized sensors and microcontrollers, such as Arduino or ESP32, to ensure affordability and scalability (Kumar et al., 2024) (Dey & Ghosh, 2023) (Aski et al., 2019). Additionally, machine learning algorithms can be applied to the collected data for predictive analytics, enabling early detection of anomalies and trends that may indicate potential health risks (Ghazaly & Jain, 2023).

The implementation of RFID-enabled IoT health check kiosks in community settings offers numerous benefits, including early detection of health issues, reduced healthcare costs, and improved health outcomes. These systems empower individuals to take a more proactive approach to their health, while also providing healthcare professionals with valuable insights for remote patient monitoring and timely intervention (Kumar et al., 2024) (Suryawanshi et al., 2023) (Ghazaly & Jain, 2023).

Moreover, by leveraging advanced sensor technologies and IoT connectivity, these systems can revolutionize community health monitoring and promote preventive care. However, despite such potential, real-world implementations highlight persistent challenges: interoperability issues, especially between RFID sensors and health information systems; data privacy concerns stemming from unsecured transmission of sensitive health details; and insufficient staff training, which significantly hampers reliability and scalability in resource-limited settings (Yadav, 2024; Profetto et al., 2022). Thus, without addressing these technical and operational gaps, RFID-enabled kiosks risk failing to deliver consistent and secure community health benefits.

RFID-enabled IoT health check kiosks should track a comprehensive set of physiological and environmental parameters, including body temperature, blood

pressure, heart rate, oxygen saturation, ECG, respiration rate, blood glucose levels, environmental factors, fall detection, and activity levels. These parameters provide a holistic view of an individual's health, enabling early detection of potential issues and improving overall well-being. By leveraging advanced sensor technologies and IoT connectivity, these systems can play a pivotal role in revolutionizing community health monitoring and promoting preventive healthcare (Xiang et al., 2022).

In direct response to the growing need for accessible and efficient healthcare services in Barangay Darapidap, this study addresses the persistent challenges of manual health screening procedures and delayed data retrieval. As a rural community in Candon City, Ilocos Sur, Darapidap continues to rely on traditional methods for checking vital signs such as height, weight, BMI, heart rate, and temperature a process that is not only time-consuming but also prone to human error. These manual procedures, compounded by overcrowding and long queues at the barangay health center, often result in delayed access to medical records and hinder timely intervention. These localized issues reflect broader national concerns, including staff shortages and the slow integration of digital health technologies.

To address these gaps, the researchers propose a localized, IoT health Check kiosk that automates vital sign monitoring and streamlines data management. This healthcare kiosk will integrate temperature, heart rate, height, weight measurements, and BMI into one device, offering a comprehensive approach to community-based health screening. The project aims to reduce the workload of healthcare workers, improve the accuracy and speed of health assessments, enable early detection of health risks, and promote preventive care in local communities. Ultimately, the importance of the study lies in delivering a low-cost, technology-driven solution to a real and growing

problem, enhancing the delivery of healthcare services in underserved areas like Barangay Darapidap.

The related studies guided of the researchers to design and develop for implementation of the kiosk by providing insights on existing systems, their strengths, and limitations. This will help the researchers turn a global and national health issue into a focused, community-level innovation. This clearly connects the background of the study to the research problem, justifying the need for the proposed project.

Conceptual Framework

A conceptual framework is a structured representation of the key variables and their presumed relationships within a study. It acts as a guide, helping researchers clarify what they aim to investigate, how different factors may influence one another, and why those connections matter. For instance, in a study examining an IoT-based health monitoring system, the framework might show how sensor accuracy (independent variable) impacts the reliability of vital signs monitoring (dependent variable), while considering user engagement and environmental factors as moderating variables. Constructing a conceptual framework typically involves identifying a clear research question, pinpointing independent and dependent variables, visualizing cause-and-effect dynamics, and acknowledging any additional influencers all of which provide a roadmap for data collection and analysis (Scribbr, 2023).

The developers designed a framework to define and justify the system to be developed, serving as a roadmap for organizing the related documentation. This framework adopts the Input-Process-Output (IPO) model, a commonly used conceptual approach that breaks down systems into three core components: inputs, processes, and outputs. The IPO model enables a clear understanding of how data or resources enter

the system, undergo specific procedures, and lead to meaningful outcomes, making it especially useful for systematic design.

By following this model, the development process flows logically from gathering user requirements and sensor inputs, through data interpretation and algorithmic processing, to generating outputs like real-time health alerts or performance reports, ensuring that each phase contributes to the clarity and coherence of the system's structure and documentation. (CIO Wiki, 2025; iSixSigma, 2025)

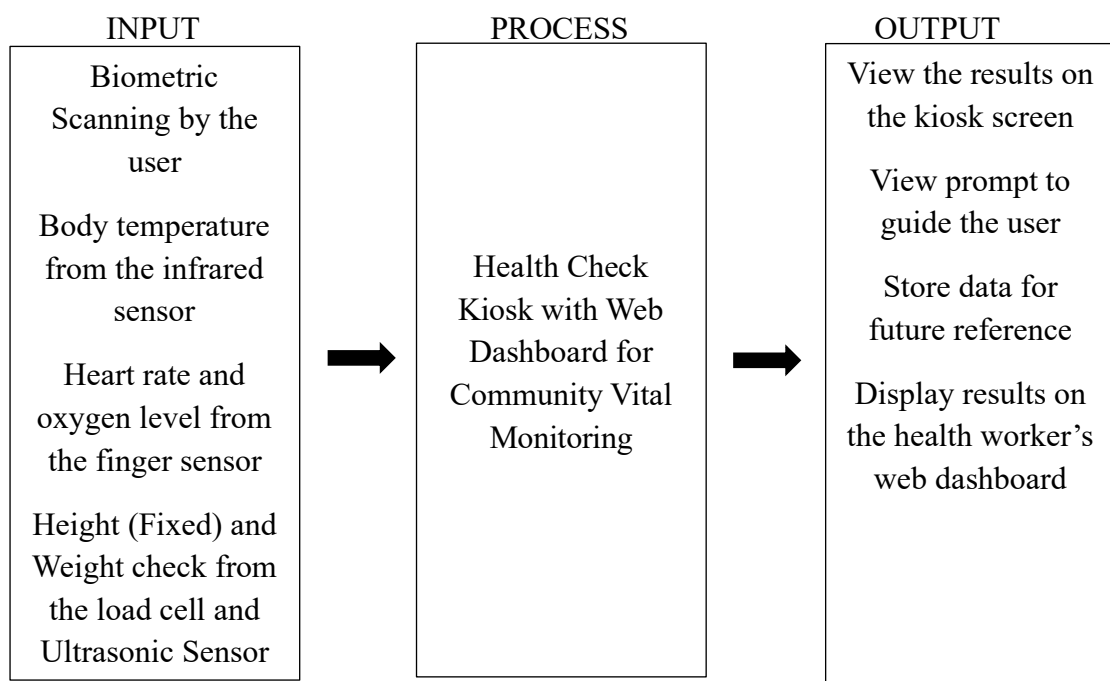


Figure 1. Research Paradigm

The research paradigm in Figure 1 follows the Input-Process-Output (IPO) model as a foundation for developing Health Check Kiosk. In the input phase, residents interact with the kiosk by biometric scanning and undergoing measurements for temperature, heart rate, oxygen level, weight, and height. These readings are then passed to the process phase, where the kiosk identifies the user, gathers data using its sensors, analyzes the results through the ESP32 microcontroller, and saves them securely in a database. Finally, in the output phase, the system presents the vital signs on the kiosk screen, available in real time through a web dashboard for authorized

personnel. This organized flow ensures accuracy, efficiency, and a more user-friendly health check experience for the community. This organized flow ensures accuracy, efficiency, and a more user-friendly health check experience for the community.

Objectives of the Study

This study aimed to design and develop the Health Check Kiosk with Web Dashboard for Community Vital Monitoring in Barangay Darapidap.

Specifically, it aimed to fulfill the following objectives:

1. To determine the information and technology requirements necessary to design and develop an IoT Health Check Kiosk with Web Dashboard for Community Vital Monitoring
2. To develop and design the features for the IoT Health Check Kiosk; and
3. To evaluate the extent of acceptability of the proposed system. In terms of:
 - a. Functionality
 - b. Reliability
 - c. Usability
 - d. Efficiency
 - e. Maintainability
 - f. Portability

Scopes and Limitation of the Study

The kiosk was developed for Barangay Darapidap to improve their local health monitoring by automating the collection and recording of basic vital signs. Residents were able to use contactless sensors to measure their height, weight, temperature, and heart rate. Health professionals or in the study's context, barangay health workers were able to view all data through web dashboard and used biometrics scanning to

identify users. The kiosk offered consistent digital records for health examinations, provide visual instruction at each step and display results in real time. It was deployed in the barangay health center and was built with ESP32, PHP, MySQL, and the necessary sensors.

The system was developed exclusively for Barangay Darapidap and will not extend to other barangays at this stage. Access to the web dashboard will be limited to authorized personnel only, and the kiosk was usable solely by registered users within the community. These constraints reflect the initial scope of implementation and are necessary to manage resources, ensure controlled deployment, and validate the system in a single, localized setting before exploring broader adoption.

Importance of the Study

This study aims to bring benefits to the following stakeholders:

To the Residents. The kiosk will provide a quick way to check vital signs such as height, weight, oxygen level, heart rate, and temperature. It will increase the convenience and accessibility of health monitoring, particularly for people living in distant locations.

To the Barangay Health Workers. Through a real-time web dashboard, the system will provide health workers a quick access to updated vital signs. To help residents respond more effectively to checks and medical emergencies, this will facilitate quicker monitoring and simpler tracking of their medical records.

To the Barangay Officials. By providing them with well-organized records of people's health data, the system will assist officials in better managing public health. This would facilitate improved preparation and prompt decision-making in the event of emergencies or local health campaigns.



To the Developers. The developers will gain practical experience in the design, construction, and testing of an Internet of Things-based health solution through this project. They will be able to use their knowledge and technical expertise while meeting actual community needs.

To the Future Developers. The study will serve as a reference for future researchers who wish to improve or adapt similar systems. It offers a foundation for exploring health technologies that are efficient, affordable, and localized.

Chapter 2

METHODOLOGY

Research Design

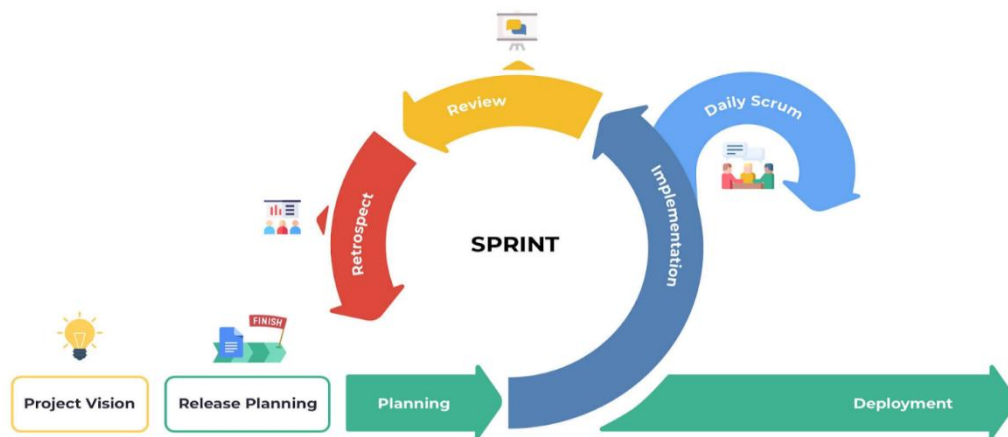
This study employed a developmental research design. This create and improve systems, tools, or programs through a process of design, testing, and refinement. It is frequently used in fields such as healthcare, technology, and education where the objective is to create workable solutions that can be evaluated and enhanced in actual environments. Developers constructed a functional prototype using this approach, get user input, and make the required modifications to improve its efficiency and usability (Clark-Plaskie et al., 2022).

One of the main advantages of this design is its flexibility. Developers can test the system while it is still being built and apply improvements based on actual user experience. It also helps connect research with real-life applications. However, this approach may take more time and effort because it involves multiple rounds of testing and revisions. It also requires close coordination with users or stakeholders, which can sometimes be difficult to manage (Clark-Plaskie et al., 2022).

This research design benefited the current study because the system being developed is intended for actual use in a community setting. Since the kiosk involves both hardware and software functions such as biometrics login, vital sign sensors, and an online health monitoring dashboard an approach that allows for hands-on evaluation and improvement is necessary. Through developmental design, the developers was able to test each component, gather feedback from health workers and residents, and apply improvements throughout the process. This helped ensure the system is functional, responsive to user needs, and ready for deployment in Barangay Darapidap.

This study adopted the Agile Development Methodology, an iterative and flexible approach to system development that focuses on delivering working components in short, manageable phases called sprints. Agile promotes continuous improvement by allowing developers to test, receive feedback, and revise the system throughout the development cycle. It emphasizes collaboration, adaptability, and early delivery of functional features, making it suitable for projects that require regular updates and user involvement.

The project adopted the Scrum framework, a flexible and iterative methodology that segments the development cycle into five core phases: Initiation, Planning and Estimation, Implementation, Review and Retrospective, and Release. This structure promotes adaptive progress, enabling the system to evolve based on continuous technical feedback and direct input from its intended users (SCRUMstudy, 2024).



Scrum Lifecycle. Sergeev, A. (2020, June 10). hygger.io. <https://hygger.io/blog/what-is-scrum-lifecycle/>

Figure 2. Scrum Lifecycle

Initiation. The Initiation Phase provides the foundational structure of the project. This stage focuses on setting clear goals, identifying target users, organizing the project team, and outlining key deliverables and potential constraints (Donato, 2023).

The developers coordinated with local officials and health workers to shape the system's direction. These consultations clarified needs related to recording vital signs through sensor technology and implementing biometric scanning. Using the input gathered, the Product owner prioritized essential features based on relevance and practicality within the community setting.

Planning and Estimation. During this phase, the developers broke down system goals into measurable user stories and assign realistic estimates each task. These stories are then grouped into parts called sprints, ensuring organized and timely execution. (Donato, 2023)

Based on initial consultations, the developers converted user needs into specific goals such as capturing height measurements or presenting health data on a local dashboard. These tasks were evaluated for complexity and scheduled strategically to balance both software and hardware development.

Implementation. This is the phase where the core components of the system are constructed. The development team completes assigned tasks in sequence while addressing issues through daily check-ins and continuous testing (Donato, 2023).

In the case of this project, each sprint focused on building modules like Biometrics authentication or sensor integration. Features were tested individually before full deployment to ensure smooth performance and early resolution of technical concerns.

Review and Retrospective. At the end of each sprint, the project team presents its output for stakeholder feedback and conducts a discussion to reflect on progress and identify improvements for the next phase (Donato, 2023).

Health workers were invited to test and evaluate system updates such as interface responsiveness and dashboard clarity. Their insights were used to drive enhancements in usability, while the team internally examined task efficiency and adjusted strategies for better collaboration.

Release. This final stage includes system completion, delivery, and support for end users. It ensures the product is operational, properly documented, and accessible to its intended audience (Donato, 2023).

Once the kiosk is fully assembled and tested, it was installed at the Barangay Darapidap health post. Staff received a complete orientation along with printed guides to support daily use and promote long-term sustainability of the system.

Project Plan

Project planning is the process of defining the objectives, scope, deliverables, timelines, resources, and activities required to complete a specific project successfully. It involves creating a comprehensive roadmap that outlines the steps needed to achieve the project's goals and objectives within the specified constraints (IdeaScale, 2024).

This planning approach supported the systematic development of the kiosk by aligning specific technical tasks with realistic timelines. It also helps the team coordinate responsibilities, anticipate challenges, and maintain focus as the system moves from design to implementation.

Figure 3 shows the project timeline for the Community Health Evaluation Center Kiosk. It displays how long each of the six phases of the Scrum Model took: Initiation, Planning, Execution, Review and Retrospective, and Release Phase. This timeline helps track the progress of the project and makes sure each step is done on time.

The project progressed through a structured sequence of phases, each justified by the practical requirements of developing a community-oriented health kiosk. The work began during the early weeks of July with the Initiation phase, which established the direction of the entire project. The team engaged directly with local officials and health workers to understand the realities of the barangay setting, allowing the goals of the system to be grounded in actual community needs. These discussions clarified functional priorities such as vital-sign sensing, and ease of use for non-technical staff. With this information, the Product Owner was able to determine which features should be delivered first, ensuring that the project moved forward with a clear and realistic scope.

As July progressed into August, the focus shifted toward Planning and Estimation. During this period the developers translated the gathered requirements into detailed user stories and technical tasks. Each task was examined for feasibility, time requirements, and interdependencies, enabling the team to organize the workload into manageable sprints. This planning effort provided a logical structure for balancing both hardware and software development, making sure that foundational components such as measurement modules, data-processing functions, and dashboard displays were prepared in an order that prevented bottlenecks later. By the end of August, the team had a complete roadmap that not only aligned with the community's priorities but also accounted for realistic technical pacing.

September marked the transition to Implementation, where the conceptual work of the earlier phases became concrete. The team focused on building essential system modules, beginning with those that required stable integration across the kiosk's hardware and software architecture. Components such as biometric authentication, vital-sign sensors, and data-transmission processes were developed and tested in

sequence to avoid compatibility issues. Daily coordination and continuous testing ensured that errors were addressed immediately, which preserved momentum and reduced the chance of major setbacks. Each completed module strengthened the system's core functionality and prepared it for the refinements that would follow.

Throughout October, the project entered Review and Retrospective. Health workers were invited to interact with the evolving system, providing practical feedback on responsiveness, clarity, and ease of use. Their observations guided adjustments that improved the interface and workflow, ensuring that the kiosk would fit smoothly into existing community health practices. The development team also used this period to reflect on their own processes, identifying areas where communication, task distribution, or scheduling could be improved in the next stages. This reflective cycle strengthened both the system's usability and the team's effectiveness.

The final weeks of the project in November were dedicated to Release. With core features completed and refinements applied, the kiosk underwent final validation to ensure stability and reliability. Once testing confirmed readiness, the system was deployed to the Barangay Darapidap health post. The team provided orientation sessions and printed guides to equip local health workers with the knowledge needed for independent operation. This final phase emphasized sustainability; by ensuring that the community could maintain and use the kiosk confidently, the project fulfilled its goal of creating a practical and long-lasting health monitoring solution.

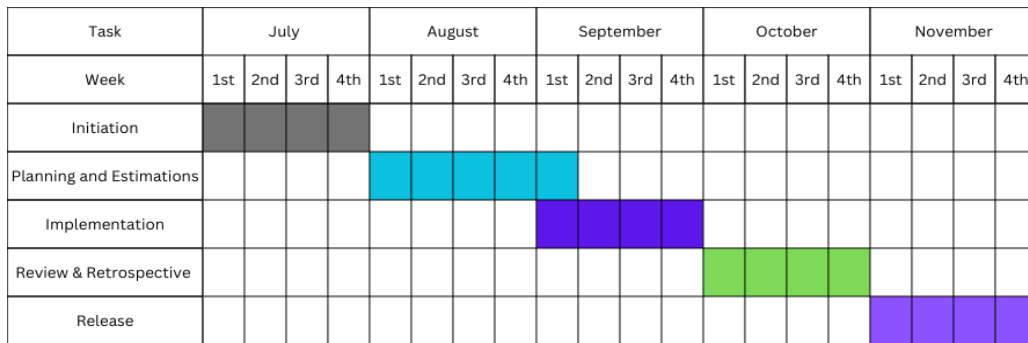


Figure 3. Gantt Chart

Project Assignments

The projects team member's roles and responsibilities within the proposed project Community Health Evaluation Center Kiosk.

Roles	Name	Responsibility
Project Manager	Vince Marc Justine Rubang	Oversee the technical team's efforts in resolving system issues and manage the planning and design team effectively.
System Analyst	Winfrey Clint Langoey, Vince Marc Justine Rubang	Responsible for building the project.
Programmer And Developer	Vince Marc Justine Rubang, Dhon Marck Hermosura	Finds and records bugs, reports problems, and helps fix errors in the program.
Quality Assurance/Tester	Melwyn Marc Ramos, Ashley Pasibe	Handle project plans, check and edit documents,

		and give updates on the project.
Documenter/Technical Writer	Dhon Marck Hermosura, Winfrey Clint Langoey, Vince Marc Justin Rubang	In charge of creating and organizing all project documents like proposals, reports, user guides, and updates, making sure they are clear and follow the rules.

Table 1. Role Requirements and Responsibility

Population and Locale of the Study

Key community health stakeholders, including as barangay officials, residents, and health workers, will make up the study's population on the IoT Health Check Kiosk with Web Dashboard in Barangay Darapidap. Their involvement will offer crucial insight into the established system's usability, dependability, and practical utility as these groups are directly involved in or impacted by the local health monitoring procedures. It is expected that health professionals in particular will be crucial to system testing, feedback gathering, and continuous assessment.

A community with active public health participation and expanding interest in technology-assisted healthcare solutions, Barangay Darapidap in Candon City, Ilocos Sur, is the study's location. The success of the kiosk in automating vital sign recording, encouraging contactless health checks, and arranging data for optimal monitoring may be assessed in a real-life situation. The study intends to evaluate how the technology

can assist community-based health initiatives by locating the system in this area. It may also be expanded to comparable barangay-level deployments throughout the region.

Table 2 shows the distribution of the selected respondents to participate in the acceptability of the proposed system.

Respondents	N
Barangay Captain	1
Brgy. Committee on Health	1
Barangay Nutrition Scholar	1
Brgy. Health Workers	12
Technical Expert	1
TOTAL	16

Table 2. Distribution of Respondents

Research Instrument

To gather essential insights for both development and evaluation, this study used several tools like interviews, informal observations, and survey questionnaires. These tools helped the developers identify how the flow of the system should work, how is the user experience, and where it can be improved.

Data was primarily gathered through in-person interviews and observation to determine the system's functional requirements and to understand the existing health monitoring process. The interviews were conducted with key community health stakeholders, including the Brgy Committee on Health and the Barangay Nurse Scholar, alongside other barangay officials and residents. This approach allowed the developers to better comprehend their specific needs and convert these requirements into essential features, such as implementing Biometrics for login and structuring the health data

dashboard. The insights derived from these sessions allowed the researchers to fine-tune the design so that it truly supports the people using it every day.

To evaluate the system's acceptability, the researchers used an evaluation sheet based on ISO 9126-2. This tool measures six acceptability indicators: Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability. (ISO, 2003). Each indicator includes statements rated on a 7-point scale, from 1 (strongly disagree) to 7 (strongly agree). Respondents included health workers, barangay officials, and residents. Their ratings identified which parts of the system are clear, easy to learn, and efficient to use. The results guided improvements before deployment.

Data Analysis

The primary statistical measure used in this research is the weighted mean, a form of average that assigns different levels of importance to individual data values through the use of corresponding weights. Unlike the arithmetic mean, the weighted mean accounts for the relative contribution of each value, making it particularly appropriate for analyzing survey responses where rating options occur at varying frequencies. It is computed by multiplying each rating by its assigned weight, summing the results, and dividing by the total sum of the weights ($\text{Weighted Mean} = \frac{\sum(w \cdot x)}{\sum w}$). This method ensures that the respondent ratings for system acceptability indicators accurately reflect overall perceptions based on scale importance and response distribution.

The data gathered from the system's acceptability survey questionnaire, which utilized an evaluation sheet based on the ISO 9126-2 standard, were analyzed using descriptive statistics. The primary statistical tool employed to interpret the results is the Weighted Mean. The Weighted Mean will be used to calculate the average rating

assigned by the respondents to each of the system's acceptability indicators (Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability).

This statistical measure is crucial for defining and interpreting the result, as the computed Weighted Mean for each indicator was mapped to a corresponding verbal scale (e.g., Highly Acceptable, Acceptable, Moderately Acceptable) to determine the overall level of acceptability of the C.H.E.C.K. system. This method provides a clear, quantitative basis for the discussion of the findings derived from the questionnaire.

Numerical Rating	Descriptive Rating
7	Strongly Agree
6	Agree
5	Slightly Agree
4	Neutral
3	Slightly Disagree
2	Disagree
1	Strongly Disagree

Table 3. Level of Agreement or Disagreement of Users Response

ISO 9126-2 defines accessibility as one of six key software quality characteristics. It breaks acceptability down into six subcomponents: functionality, Reliability, Usability, Efficiency, Maintainability, and Portability. These indicators help determine how easy and pleasant a system is for users to interact with.

Chapter 3

Results and Discussion

In this chapter, the study findings were outlined, covering the current process of vital signs monitoring in the community, the features of the developed C.H.E.C.K system, and the level of acceptability of the system.

Findings

The following are findings in relation to the interview that led to the conduct of the study:

The Information Technology Requirements for C.H.E.C.K Kiosk

The current vital signs collection process in the community was determined through direct observation and interviews with barangay health workers. Based on health staff accounts, residents' height, weight for BMI, pulse rate, and temperature are measured manually using basic medical equipment during scheduled health events or house-to-house visits. Measurements are recorded by hand in logbooks, as no centralized or standardized recording system is in place. When inconsistencies occur, health workers repeat the measurement process to verify accuracy before submitting reports to the rural health unit.

Development of the IoT System

This encompasses the design, integration, and implementation of both hardware and software components.

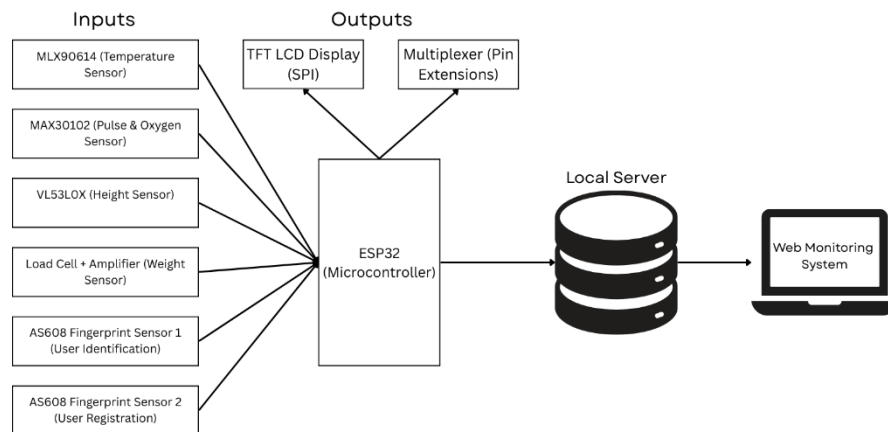


Figure 4. Block Diagram of the CHECK System

The system utilizes an MLX90614 temperature sensor, MAX30102 pulse and oxygen sensor, VL53L0X height sensor, and a load cell with amplifier to measure residents' vital signs. In addition, two AS608 fingerprint sensors are employed for user registration and authentication. All sensor data are processed by the ESP32 microcontroller, which also manages biometric verification. The processed data are then transmitted to a local server for storage and can be accessed through a web monitoring system, allowing administrators to view and analyze residents' health records in real time.

Hardware Components

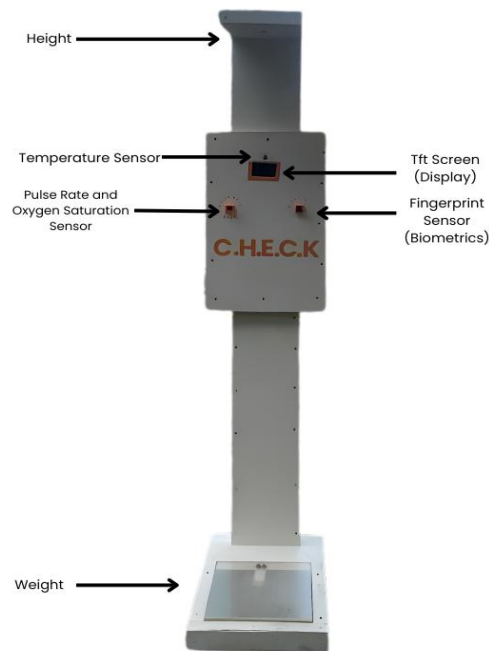


Plate 8. Functional Design of CHECK System

The system is primarily powered by a direct electricity connection, ensuring continuous operation of the kiosk. The ESP32 microcontroller regulates the distribution of power to the connected hardware components, including the fingerprint sensors, temperature sensor, pulse and oxygen saturation sensor, height sensor, weight sensor, and TFT display. Once the kiosk is plugged in, electricity flows through the device, supplying stable power to all modules.

The kiosk enclosure is designed with two main sections: the biometric authentication unit and the vital signs measurement unit. The biometric unit houses the AS608 fingerprint sensors for resident registration and authentication. After successful authentication, the vital signs unit activates, measuring the resident's temperature (MLX90614), pulse and oxygen level (MAX30102), height (VL53L0X), and weight (load cell with amplifier). The TFT LCD screen displays the readings instantly for verification.

Upon startup, the ESP32 establishes communication with the local server. The TFT display can also serve as a status indicator for connectivity and operational readiness. Sensor readings are processed by the ESP32 and then stored in the local server; each record tagged with the authenticated resident's identity. The web monitoring system retrieves the stored data in real time, providing administrators with statistical reports and resident health records.

This setup ensures that the kiosk operates efficiently with a stable electricity supply, delivering accurate, automated, and secure vital signs monitoring for the community.



Figure 5. ESP32 Microcontroller

Microcontroller. serves as the central processing unit of the C.H.E.C.K. kiosk, orchestrating all system operations with precision and efficiency. It manages sensor inputs, handles biometric authentication through the AS608 fingerprint module, processes vital sign data, and communicates with the local or cloud-based server for storage and dashboard updates. This dual-core microcontroller is powerful enough to multitask reading sensors, updating the display, and transmitting data simultaneously making it ideal for a responsive, real-time health kiosk.



Figure 6. AS608 Fingerprint Sensor.

Fingerprint Sensor. This biometric sensor plays a critical role in ensuring that each health scan is securely linked to the correct resident. It allows for both enrollment and authentication, storing fingerprint templates onboard and verifying identity before any vital signs are measured. This step guarantees that all health data is personalized and traceable, which is essential for accountability and accurate recordkeeping.



Figure 7. MLX90614 Temperature Sensor.

Temperature Sensor. The MLX90614 infrared sensor provides a fast, contactless solution. It reads skin temperature within seconds, minimizing wait time and reducing the risk of cross-contamination, an important feature in public or high-traffic settings. Alongside it, the MAX30102 sensor captures both pulse rate and oxygen saturation using optical methods.

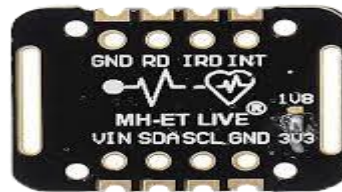


Figure 8. MAX30102 Pulse oximeter and Heart-rate sensor.

Pulse and Heart Rate Sensor. The MAX30102 is a dual-function optical sensor that measures both pulse rate and blood oxygen saturation (SpO_2). It operates by emitting red and infrared light into the skin and analyzing the reflected signals to detect changes in blood volume. These changes correspond to the heartbeat and oxygen levels, providing essential cardiovascular and respiratory data. Once a resident is authenticated, the sensor performs a quick, non-invasive scan, allowing the system to capture vital signs in real time. Its compact design and high sensitivity make it ideal for embedded health systems, contributing to early detection of silent conditions such as hypoxia or arrhythmia.

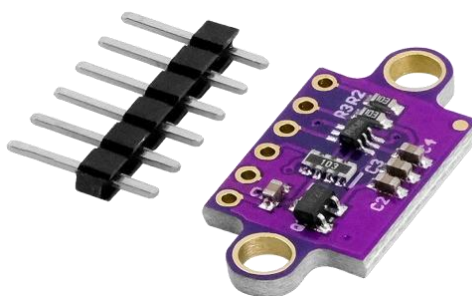


Figure 9. VL53L0X (Time of Flight) Distance Sensor

Height Sensor. The VL53L0X is a laser-based time-of-flight sensor used to measure the resident's height. It calculates the distance between the sensor and the top of the head with high precision and speed. Its compact form factor allows for seamless integration into the kiosk structure, and its non-contact operation ensures hygienic and efficient measurements. When paired with weight data, the height reading enables the system to compute Body Mass Index (BMI), providing a more complete picture of the resident's physical health.

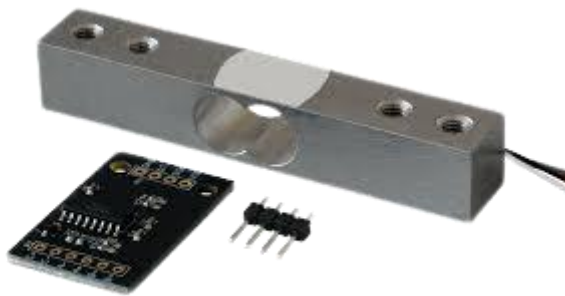


Figure 10. Load Cell (200kg) + Amplifier

Load Cell with Amplifier. This strain gauge load cell, combined with the HX711 amplifier, measures the resident's weight with high accuracy and stability. The amplifier boosts the low-voltage signal from the load cell, allowing the ESP32 to interpret the data reliably. The sensor supports up to 200kg and maintains consistent readings even in varied environmental conditions. Weight data is essential for BMI calculation and general health assessment, especially in nutrition-sensitive communities.



Figure 11. TFT LCD Display

TFT LCD Display. The TFT LCD display serves as the primary user interface of the kiosk, providing real-time feedback and guiding residents through each step of the scanning process. It displays prompts during fingerprint authentication, shows vital sign results after successful scans, and communicates system status to both users and administrators. This visual layer enhances usability, ensuring that the kiosk remains accessible and intuitive for all residents, including those with limited technical experience.

Features of the developed System

The administrators can utilize the system to effectively manage the collection and monitoring of residents' vital signs, to track individual health records, and to interpret health data. It features an intuitive dashboard with reporting capabilities, providing visual and statistical information on temperature, pulse rate, height, weight. Administrators can define manage resident profiles with biometric registration. The system enhances community healthcare by integrating biometric authentication, IoT-based sensors, and automated recording, ensuring efficiency, accuracy, and sustainability in vital signs monitoring.

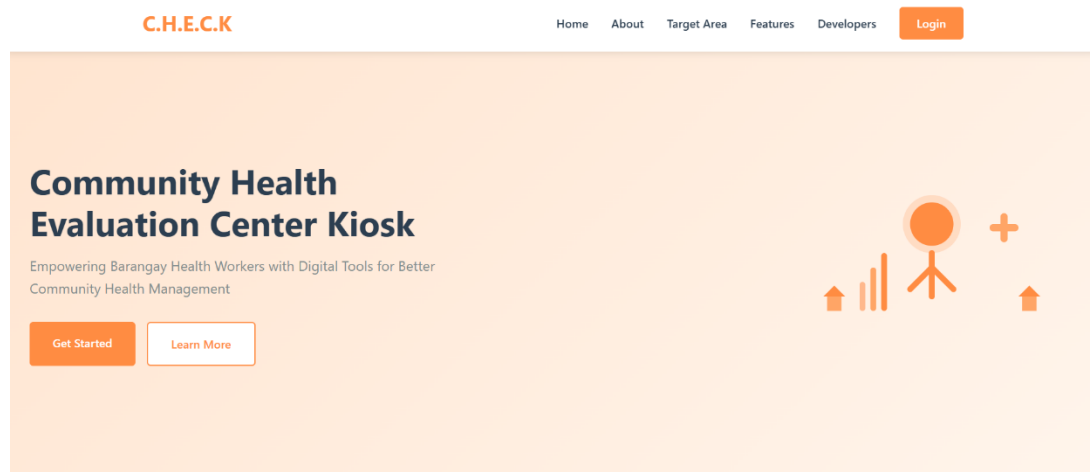


Plate 1. Landing Page

The landing page serves as the system's entry point, offering a clean and welcoming interface for administrators. It introduces the purpose of the C.H.E.C.K. kiosk and provides quick access to core modules such as resident registration, vital records, and system support. Visually, it sets the tone for usability and professionalism, often featuring the system logo, a brief description, and login credentials. This page reinforces trust and clarity, especially for first-time users or barangay health workers unfamiliar with digital systems.

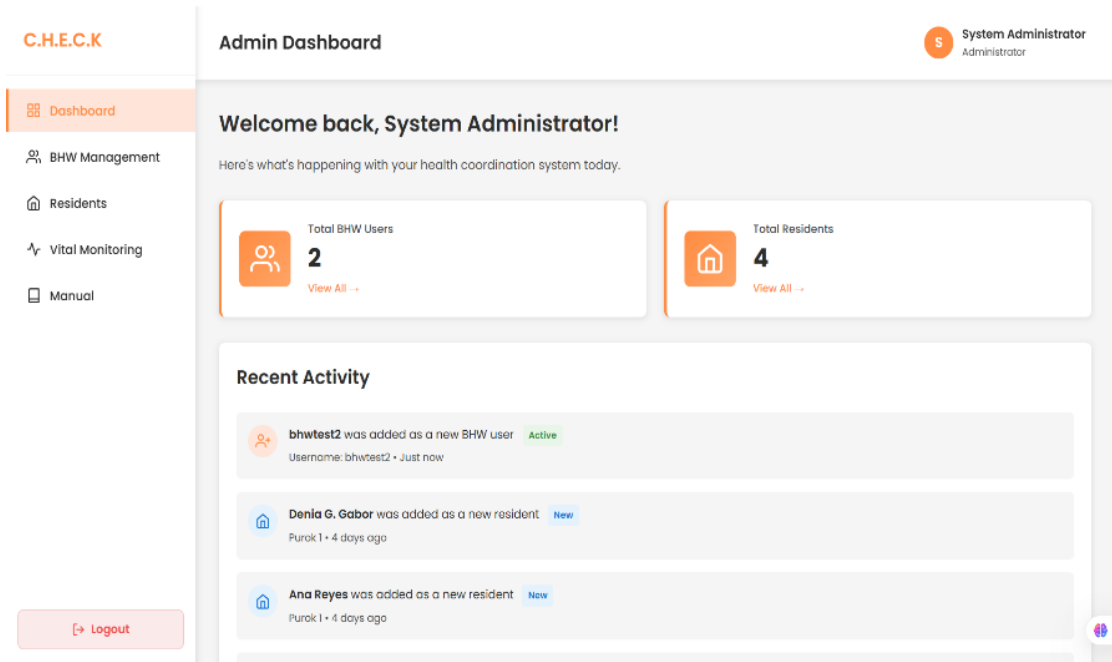


Plate 2. Admin Dashboard

The admin dashboard is the control center of the system. It presents an overview of key metrics such as the number of registered residents, through visual charts and summaries. This dashboard empowers administrators to monitor health in real time. It also includes navigation to submodules like Residents, Vital Records, and Support, streamlining workflow and reducing training time. The dashboard's intuitive layout ensures that even non-technical users can operate the system confidently

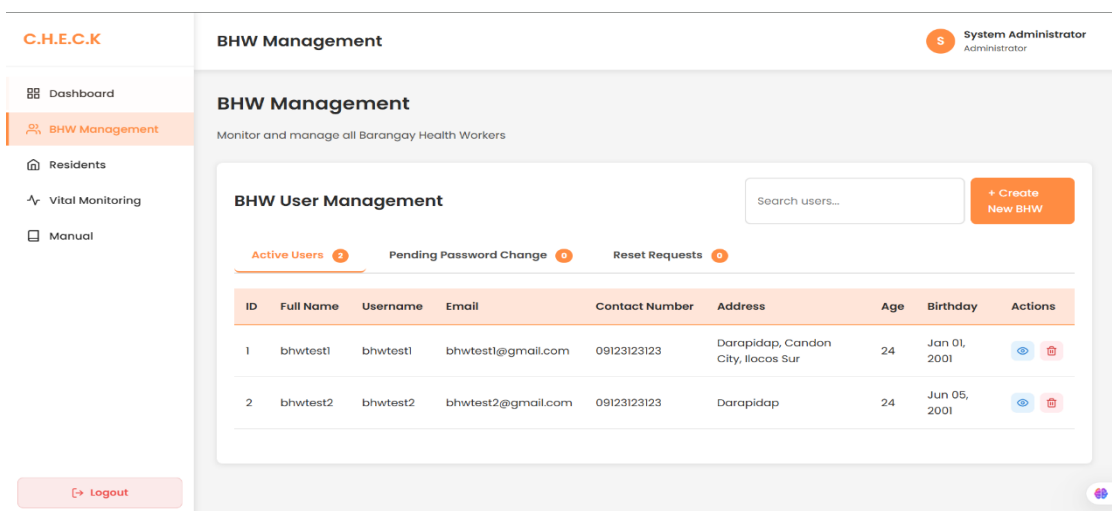


Plate 3. Admin – BHW Management

The BHW Management module enables administrators to create, manage, and monitor Barangay Health Worker (BHW) user accounts. The administrator can add new BHWs by providing details such as full name, username, password, contact information. Editing and deleting accounts are also supported, ensuring that only authorized and active BHWs can access the system. This feature strengthens data security and accountability by ensuring that each vital record can be traced back to an authorized health worker. The interface provides a clear list of BHW users, along with tools for search and account management, promoting organized and secure operations within the health system.

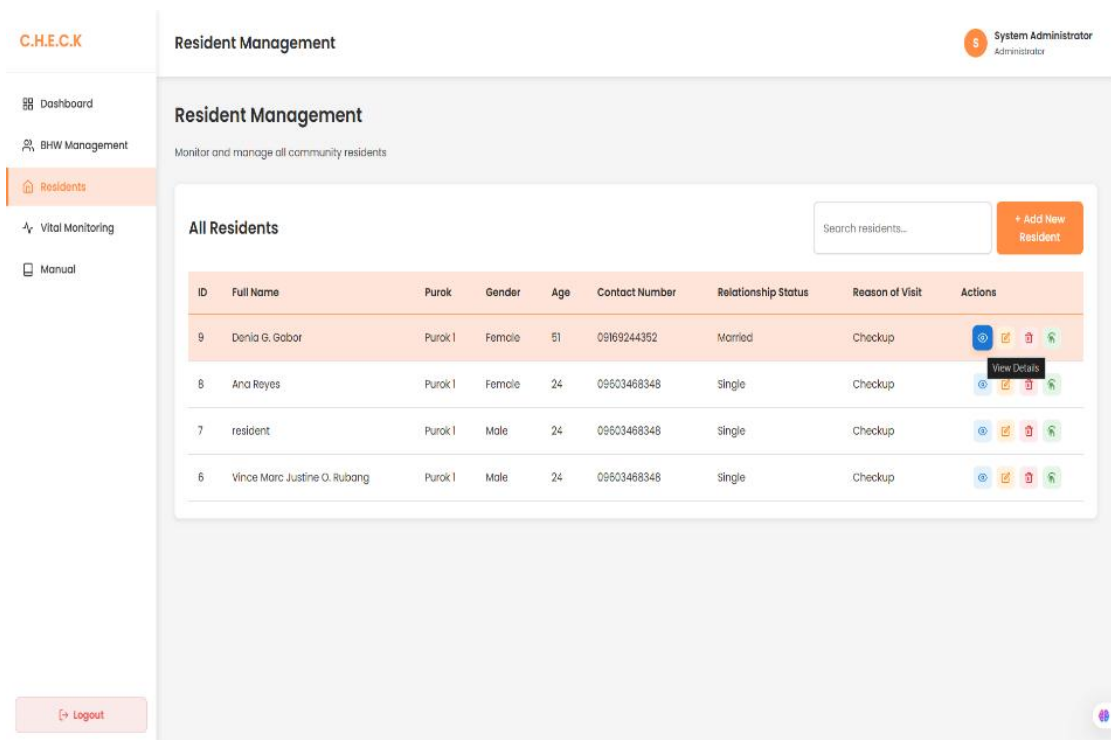
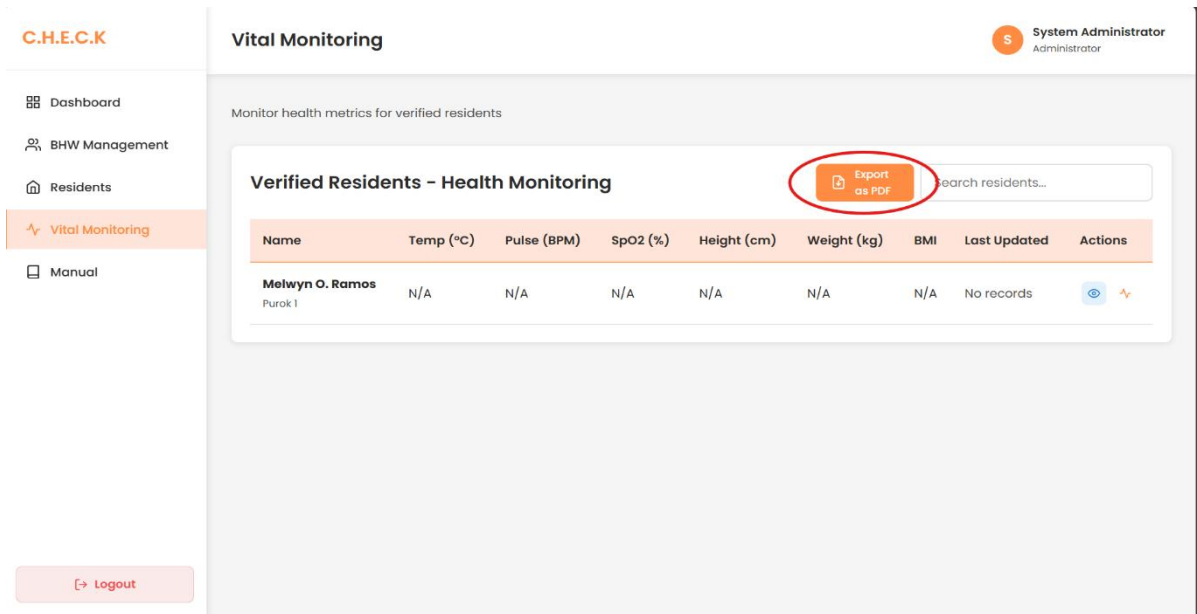


Plate 4. Admin - Residents' Management

This module allows administrators to register and manage resident profiles. It includes a form for entering personal details such as name, age, purok, and reason for visit, followed by biometric enrollment using the AS608 fingerprint sensor. Each resident is assigned a unique fingerprint ID, ensuring that future scans are securely linked to their profile. The interface also supports editing, deleting, and searching for

residents, making it easy to update records or verify identity. This plate is crucial for maintaining accurate, traceable health data and preventing duplicate entries.



Name	Temp (°C)	Pulse (BPM)	SpO2 (%)	Height (cm)	Weight (kg)	BMI	Last Updated	Actions
Melwyn O. Ramos Purok 1	N/A	N/A	N/A	N/A	N/A	N/A	No records	 

Plate 5. Admin – Vital Monitoring

The Vital Monitoring page displays all health scans linked to registered residents. After successful fingerprint authentication, the system captures temperature, pulse rate, oxygen saturation, height, weight. These readings are stored with timestamps and user identifiers, allowing administrators to track individual health over time. The interface includes filters by name, purok, reason for visit, and may offer export options for reporting. This plate transforms raw sensor data into actionable insights, supporting early detection and community health planning.

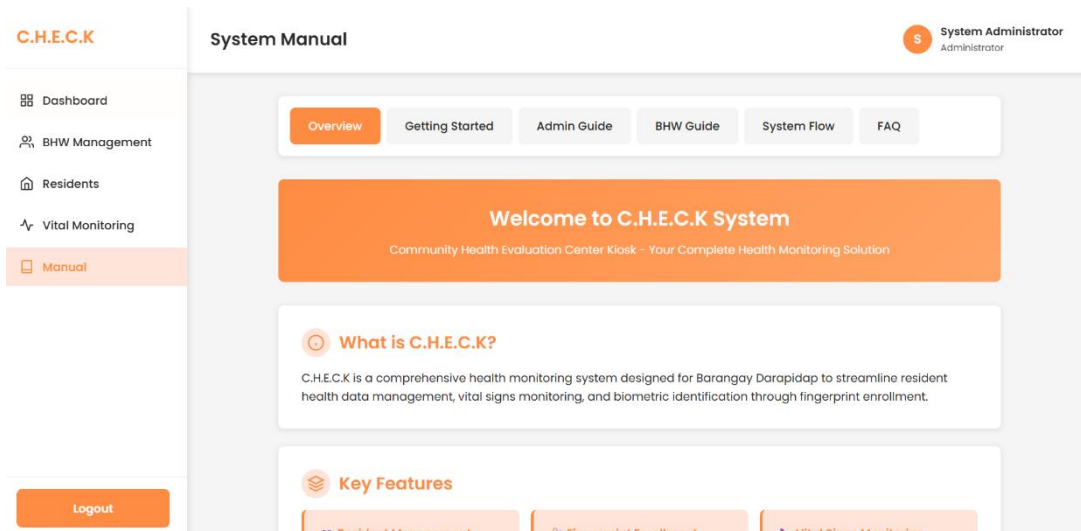


Plate 6. Admin – Support/Manual

This section provides guidance for system use, troubleshooting, and maintenance. It includes onboarding instructions, sensor calibration tips, and FAQs tailored to local deployment. Visual aids and culturally adaptive language help ensure that barangay health workers can operate the kiosk independently. This plate reinforces sustainability by reducing reliance on external technical support and empowering local administrators to manage the system confidently

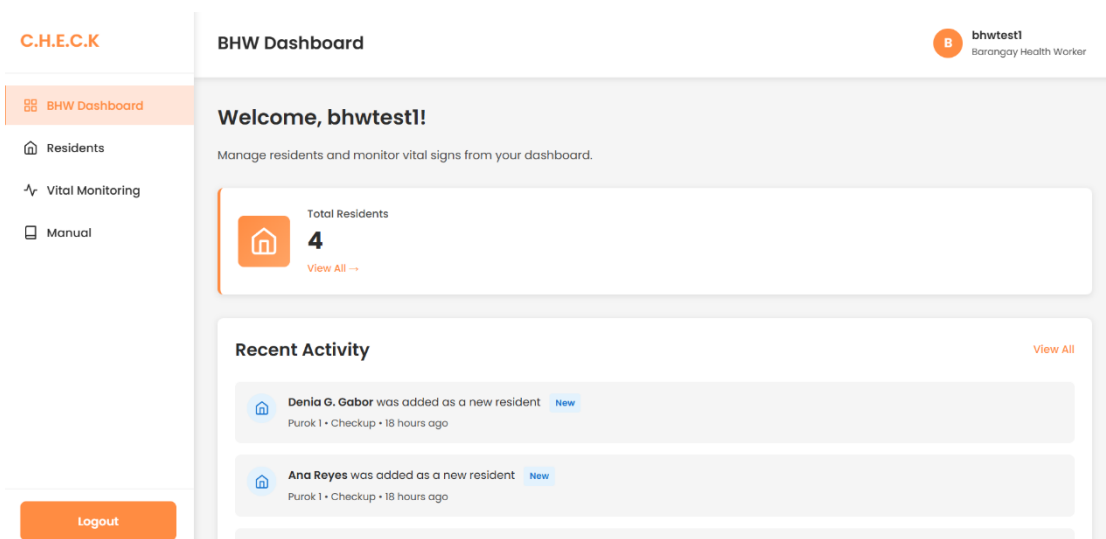
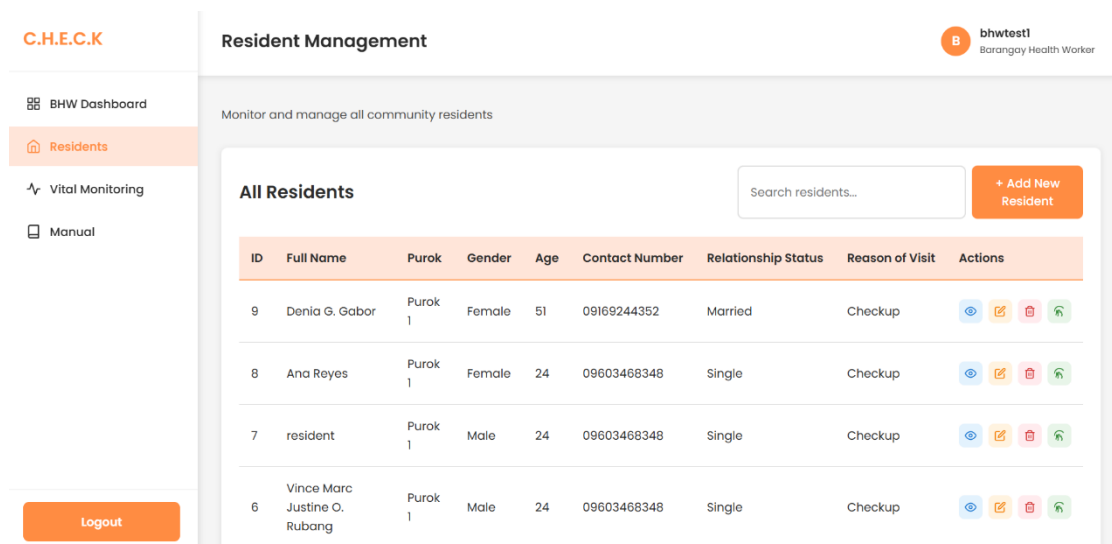


Plate 7. BHW Dashboard

The BHW Dashboard serves as the primary workspace for Barangay Health Workers. Upon logging in, BHW users can search for residents using name, fingerprint, or ID and record their vital signs, including temperature, pulse rate, oxygen saturation, height, and weight. The dashboard also allows them to view a resident's previous records for follow-up visits and submit new data to the system in real time. Unlike the admin dashboard, this interface excludes the BHW Management module to maintain security and role separation. Its simplified design ensures that BHWs can focus on efficient data entry and health monitoring, especially during field operations.



ID	Full Name	Purok	Gender	Age	Contact Number	Relationship Status	Reason of Visit	Actions
9	Denia G. Gabor	Purok 1	Female	51	09169244352	Married	Checkup	   
8	Ana Reyes	Purok 1	Female	24	09603468348	Single	Checkup	   
7	resident	Purok 1	Male	24	09603468348	Single	Checkup	   
6	Vince Marc Justine O. Rubang	Purok 1	Male	24	09603468348	Single	Checkup	   

Plate 8. BHW Residents Management

This module allows Barangay Health Workers to search for and view existing resident profiles. Each resident can be located through name, ID, or biometric fingerprint verification. Once verified, the BHW can review personal details such as name, age, address, and reason for visit. The system ensures accuracy by preventing BHW users from editing or deleting records maintaining data integrity and centralized control under the administrator. This feature enables BHWs to efficiently manage residents during medical visits and community health assessments while ensuring that each entry remains secure and consistent with the main database.

C.H.E.C.K

BHW Dashboard

Residents

Vital Monitoring

Manual

Logout

Vital Monitoring

Monitor health metrics for verified residents

Verified Residents - Health Monitoring

Export as PDF

Search residents...

Name	Temp (°C)	Pulse (BPM)	SpO2 (%)	Height (cm)	Weight (kg)	BMI	Last Updated	Actions
Ana Reyes Purok 1	30.6	N/A	N/A	N/A	N/A	N/A	Nov 18, 2025 09:35 AM	 
Denia G. Gabor Purok 1	32.9	N/A	N/A	N/A	N/A	N/A	Nov 18, 2025 09:54 AM	 
resident Purok 1	33.4	N/A	N/A	N/A	N/A	N/A	Nov 18, 2025 09:21 AM	 
Vince Marc Justine O. Rubang Purok 1	36.2	83	90.0	N/A	N/A	N/A	Nov 17, 2025 09:14 PM	 

Plate 9. BHW – Vital Monitoring

The Vital Records page enables BHWs to capture and record residents' health data directly from the sensors after successful biometric authentication. Data such as temperature, pulse rate, oxygen saturation, height, and weight are automatically stored in the system, tagged with the resident's identity and timestamp. BHWs can also view previous vital sign histories to track health progress or identify unusual readings that may require further attention. This feature streamlines the health data collection process and ensures that all records are updated in real time, improving the efficiency and accuracy of barangay-level healthcare monitoring.

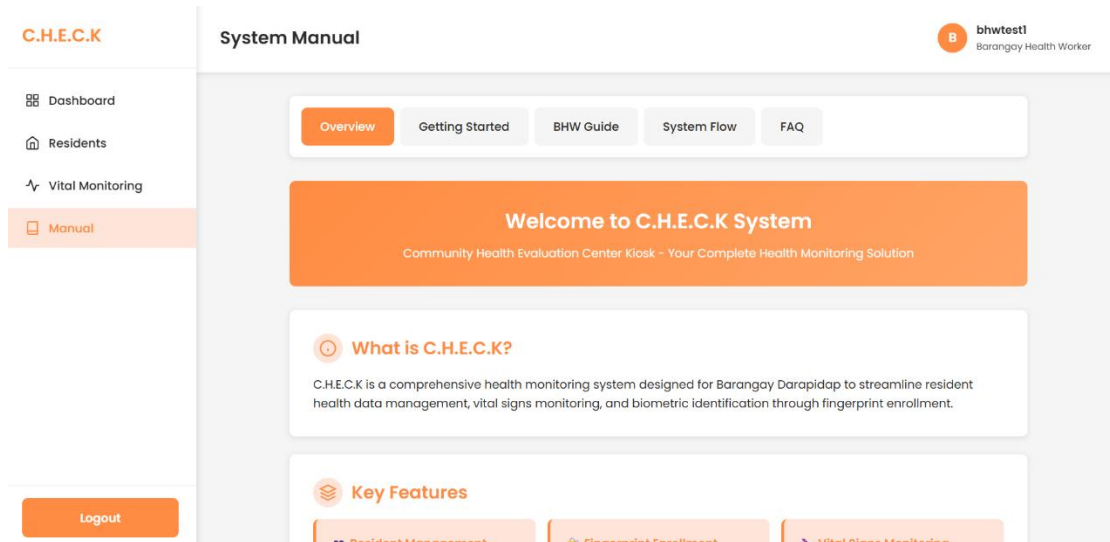


Plate 10. BHW – Support/Manual

The Support or Manual section serves as a guide for BHW users in operating the system. It provides step-by-step instructions on how to use the resident management and vital records modules, including troubleshooting tips for sensor connection, biometric scanning, and data uploading. The manual uses clear and simple language suitable for field users, ensuring that health workers can operate the system confidently even without technical expertise. This feature promotes self-reliance, reduces errors during operation, and reinforces the system's long-term sustainability at the barangay level.

Extent of Acceptability of the Proposed Application

To determine the acceptability of the C.H.E.C.K Kiosk and address Objective Three (3) of this study, an evaluation instrument based on the ISO/IEC 9126-2 standard was employed (ISO/IEC, 2001). This standard specifies external quality metrics for software and hardware products, focusing on six indicators: Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability.

On the Functionality part, which assesses the system's ability to perform its intended tasks accurately, received an overall mean score of 6.79, corresponding to a "Strongly Agree" rating. The highest mean of 6.88 was observed for the statement, "The system performs all intended functions correctly". This highlights strong confidence that the kiosk successfully fulfills its core health monitoring mission. However, the lowest mean, 6.75 (Agree), was shared by two items: "The system provides accurate results based on user input" and "The system's features meet the user's needs". This indicates that while the system is highly functional, a minor gap exists between the expected scope of features and the final design, implying potential limitations or features that users expected but were not included, thus affecting the perceived full alignment with needs.

Sub-Indicators	Mean	Descriptive Rating
1. The system performs all intended functions correctly.	6.88	Strongly Agree
2. The system provides accurate results based on user input.	6.75	Strongly Agree
3. The system's features meet the user's needs.	6.75	Strongl Agree
Overall Rating	6.79	Strongly Agree

Table 4. Functionality of the System

For the Reliability part, pertaining to the system's stability and consistent performance, achieved an overall mean score of 6.33, rated as "Agree". The highest mean score of 6.38 was shared by the statements, "The system works without unexpected crashes or errors" and "The system gives consistent results during use". These scores affirm that the system generally maintains a stable operating environment. Conversely, the expectation that "The system can recover easily after errors or failures" received the lowest mean of 6.25 (Agree). This suggests that although the system rarely fails, when an error occurs, the process of recovery or resuming operation might not be as seamless or fast as users would prefer, pointing to a need for more robust exception handling.

Sub-Indicators	Mean	Descriptive Rating
1. The system works without unexpected crashes or errors.	6.38	Agree
2. The system gives consistent results during use.	6.38	Agree
3. The system can recover easily after errors or failures.	6.25	Agree
Overall Rating	6.33	Agree

Table 5. Reliability of the System

With regards to Usability part, which measures the ease and efficiency of the system's interface, achieved an overall mean score of 6.79, corresponding to a "Strongly Agree" rating. The highest mean of 6.88 was given to the item, "The system is easy to use and understand". This outcome confirms the success of the system's design in providing an intuitive and easily navigable experience for community health stakeholders. The lowest mean, 6.69 (Agree), was for the item "The system helps users' complete tasks quickly and correctly". This finding, while positive, suggests that although the system is easy to figure out, there is room to refine the interaction flow to further optimize task completion speed and efficiency.

Sub-Indicators	Mean	Descriptive Rating
1. The system is easy to use and understand.	6.88	Strongly Agree
2. The system's interface is user-friendly.	6.81	Strongly Agree
3. The system helps users' complete tasks quickly and correctly.	6.69	Strongly Agree
Overall Rating	6.79	Strongly Agree

Table 6. Usability of the System

As to Efficiency part, which measures performance relative to time and resources, achieved the highest overall mean score among all six indicators at 6.88, resulting in a "Strongly Agree" rating. Both sub-indicators "The system loads and responds quickly" and "The system performs tasks without unnecessary delays" received an identical mean of 6.88. This collective highest score strongly indicates that the system operates with optimal speed and minimal latency, which is essential for ensuring fast and effective patient throughput in a time-sensitive health screening environment.

Sub-Indicators	Mean	Descriptive Rating
1. The system loads and responds quickly.	6.88	Strongly Agree
2. The system performs tasks without unnecessary delays.	6.88	Strongly Agree
Overall Rating	6.88	Strongly Agree

Table 7. Efficiency of the System

On the Maintainability part, which assesses the ease of modification and correction, recorded an overall mean score of 6.75, resulting in an "Agree" rating. All three sub-indicators under this characteristic "The system is easy to update or fix when needed," "The system's code or structure is well organized," and "The system allows adding new features easily" received an identical mean of 6.75 (Agree). This uniformity suggests that evaluators perceive the system as having a clear, well-structured design that facilitates easy ongoing technical support, maintenance, and future expansion.

Sub-Indicators	Mean	Descriptive Rating
1. The system is easy to update or fix when needed.	6.75	Strongly Agree
2. The system's code or structure is well organized.	6.75	Strongly Agree
3. The system allows adding new features easily.	6.75	Strongly Agree
Overall Rating	6.75	Strongly Agree

Table 8. Maintainability of the System

With regards Portability part, defined as the ease with which the system can be adapted or moved to a different environment, attained an overall mean score of 6.77, resulting in an "Agree" rating. The highest mean of 6.81 (Strongly Agree) was for the item, "The system can run on different devices or platforms". This finding suggests that the system's architectural design is flexible. Conversely, the lowest mean, 6.75 (Agree), was shared by the statements "The system works properly after being moved to another environment" and "The system needs only small changes to work on other systems". These scores collectively imply that while the software is generally platform-independent, minor configuration or calibration efforts may still be necessary when physically relocating the kiosk or integrating it with disparate health systems.

Sub-Indicators	Mean	Descriptive Rating
1. The system is easy to update or fix when needed.	6.81	Strongly Agree
2. The system's code or structure is well organized.	6.75	Strongly Agree
3. The system allows adding new features easily.	6.75	Strongly Agree
Overall Rating	6.77	Strongly Agree

Table 9. Portability of the System

The summary of evaluation is shown in Table 10, with a grand mean of 6.72 which corresponds to “Strongly Agree”. The system’s greatest strengths lie in areas directly affecting user experience and productivity: Efficiency (Mean: 6.88), Functionality (Mean: 6.79), and Usability (Mean: 6.79). Efficiency was the highest-rated indicator, with evaluators confirming the system's ability to operate with optimal speed and minimal latency (Mean: 6.88 for quick response and no unnecessary delays). Similarly, high scores for Functionality confirmed that the kiosk successfully performs all intended health monitoring tasks (Mean: 6.88), while Usability results confirmed the system is "easy to use and understand" (Mean: 6.88), validating the success of the intuitive interface design.

However, the evaluation also identified a critical area for future refinement: Reliability (Overall Mean: 6.33, Agree). This was the lowest-rated characteristic, driven by the lowest sub-indicator score of 6.25 for the item "The system can recover easily after errors or failures." This finding suggests that while the system is generally stable,

the mechanism for error handling and recovery needs to be made more seamless and robust. Furthermore, a minor gap in perceived completeness was noted under Functionality, where "The system's features meet the user's needs" received a low mean of 6.75 (Agree), indicating a slight discrepancy between user expectations and the final feature set. The overall results affirm that the system is a high-quality, viable solution, with a specific focus on enhancing its operational endurance in challenging field conditions.

Indicators	Mean	Descriptive Rating
1. Functionality	6.79	Strongly Agree
2. Reliability	6.33	Agree
3. Usability	6.79	Strongly Agree
4. Efficiency	6.88	Strongly Agree
5. Maintainability	6.75	Strongly Agree
6. Portability	6.77	Strongly Agree
Overall Rating	6.72	Strongly Agree

Table 10. Summary of Evaluation of the Developed System

Conclusion

In reference to the findings, the researchers conclude that:

1. Based on observations of current practices in local health units, the Barangay Health Workers (BHWs) and residents historically relied on manual and paper-based processes for recording vital signs (e.g., temperature, pulse, height, and weight). This manual system inherently

resulted in data transcription errors, time-consuming reporting, and inefficient management of community health records.

2. The researchers successfully developed and implemented the C.H.E.C.K (Community Health Evaluation Center Kiosk), an IoT-based system designed to automate and streamline these procedures. The Kiosk integrates multiple non-invasive sensors to automatically measure and record key vital signs, securely transferring the data to a centralized, real-time web dashboard for access by authorized health personnel.
3. The evaluation of the C.H.E.C.K Kiosk indicates a highly positive acceptance from its intended users. The system was recognized for its efficiency, strong functionality, and intuitive usability. The overall feedback affirms that the C.H.E.C.K Kiosk is a valuable, effective, and necessary technological intervention for enhancing the data management and initial screening capabilities of the local barangay health center.

Recommendations

Based on the findings and limitations identified during the development and evaluation of the system, the following recommendations are proposed to further enhance its accuracy, reliability, and clinical relevance:

1. To ensure accuracy across all age groups (pediatric, adult, and elderly), the height measurement system must be upgraded from a fixed sensor to an adjustable mechanism capable of precisely measuring a wider demographic range. This is critical for reliable Body Mass Index (BMI) calculation.
2. A technical review of the sensor output must be conducted to verify the exact physiological parameter being measured (e.g., differentiating between Heart

Rate derived from electrical signals and Pulse Rate derived from blood flow/volume). The system interface and documentation must then use the correct, standardized clinical term to avoid ambiguity for BHWs and clinicians.

3. To mitigate measurement errors caused by user movement and ensure the stability of the load cell readings, the physical platform for the weight scale must be structurally reinforced to provide maximum rigidity and stability.
4. The web dashboard and patient report generation features should be updated to incorporate and apply recognized BMI bracketing standards (e.g., WHO, DOH standards) that are specific to different demographic factors such as age and gender.

GLOSSARY

Agile Development Methodology - An iterative and flexible approach to system development that focuses on delivering working components in short, manageable phases called sprints, promoting continuous improvement.

BHW - Barangay Health Worker. Key community health personnel who use the C.H.E.C.K system to view, track, and manage residents' health data through the web dashboard.

Biometric Scanning - The use of unique physical characteristics, specifically fingerprint sensors (AS608) in the Kiosk, for reliable user authentication and resident identification

BMI - Body Mass Index. A metric calculated from a person's height and weight used by the system as a screening tool for general health assessment.

Developmental Research Design - A research approach employed to create and improve systems, tools, or programs through a process of continuous design, testing, and refinement.

Efficiency - A quality characteristic of the system (per ISO 9126-2) referring to the performance relative to the number of resources used, such as response time and processing speed.

ESP32 Microcontroller - The core processing unit of the Kiosk that gathers and analyzes data from the sensors, manages biometric verification, and saves the results securely in the database.

Functionality - A quality characteristic (per ISO 9126-2) that defines the capability of the software product to provide the required functions that meet user needs.

Input-Process-Output Model. The conceptual framework used for the Kiosk development, which logically breaks down the system into the data collected (Input), the processing of that data (Process), and the resulting information (Output).

IoT - Internet of Things. A system that leverages technology to wirelessly capture and transmit health data from physical sensors to a cloud platform for storage and analysis

ISO 9126-2 - An international standard used to evaluate the system's acceptability, measuring six indicators of software quality including Functionality, Reliability, and Usability.

Load Cell - A sensor component, typically combined with an amplifier, used in the Kiosk to accurately and stably measure the resident's weight.

Maintainability - A quality characteristic (per ISO 9126-2) relating to the effort needed to modify the system, including making corrections, improvements, or adapting to changes.

MAX30102 - A pulse oximeter sensor used in the Kiosk to measure the user's heart rate and oxygen level (SpO2).

MLX90614 - An infrared sensor used in the Kiosk to measure the user's body temperature.

Portability - A quality characteristic (per ISO 9126-2) describing the software's ability to be transferred from one environment to another.

Reliability - A quality characteristic (per ISO 9126-2) defining the ability of the system to maintain a specified level of performance under stated conditions for a specified period.

Scrum Framework - The specific Agile methodology adopted, which segments the development cycle into five phases: Initiation, Planning, Implementation, Review, and Release.

TFT LCD Display - The primary User Interface (UI) screen of the Kiosk, providing real-time feedback, guiding prompts, and displaying vital sign results to residents.

Usability - A quality characteristic (per ISO 9126-2) that reflects the effort needed for system use and the user's subjective assessment of the system.

Vital Signs - Essential physiological measurements (temperature, heart rate, oxygen level, height, and weight) collected by the Kiosk to assess an individual's health status.

VL53L0X - A Time of Flight Sensor used in the Kiosk to measure the user's height.

Web Dashboard - The centralized, password-protected web interface provided for authorized BHWs to access, view, and track updated vital sign data and medical records in real time.

References

- Aski, V. J. (2019, January 1). *IoT enabled ubiquitous healthcare data acquisition and monitoring system for personal and medical usage powered by cloud application: An architectural overview*. SciSpace. Retrieved June 24, 2025, from <https://scispace.com/papers/iot-enabled-ubiquitous-healthcare-data-acquisition-and-spzljp6ut7>
- Aski, V. J., Sonawane, S. S., & Soni, U. (2019). IoT enabled ubiquitous healthcare data acquisition and monitoring system for personal and medical usage powered by cloud application: An architectural overview. In J. Kalita, V. Balas, S. Borah, & R. Pradhan (Eds.), *Recent developments in machine learning and data analytics* (Vol. 740). Springer, Singapore.
https://doi.org/10.1007/978-981-13-1280-9_1
- Bellio, J. (2024, December 19). *System usability scale (SUS) practical guide for 2025*. UXtweak. Retrieved June 30, 2025, from <https://blog.uxtweak.com/system-usability-scale/>
- Clark-Plaskie, M. (n.d.). *Developmental research designs – Individual and family development, health, and well-being*. Iowa State University Digital Press. Retrieved June 22, 2025, from <https://iastate.pressbooks.pub/individualfamilydevelopment/chapter/developmental-research-designs/>
- Dey, N., & Ghosh, P. (2023). An IoT-enabled vital cardiac parameter monitoring system on real-time basis. In *Proceedings of International Conference on Data, Electronics and Computing Algorithms for Intelligent Systems* (pp. 201–220). Springer Nature Singapore. https://doi.org/10.1007/978-981-99-1509-5_19

Donato, H. (2023, January 3). *What are the phases of Scrum?* Workamajig. Retrieved June 27, 2025, from <https://www.workamajig.com/blog/scrum-methodology-guide/scrum-phases>

EDP Sciences. (2024, November 14). IoT based remote patient health monitoring system. *E3S Conferences*. Retrieved June 24, 2026, from https://www.e3s-conferences.org/articles/e3sconf/abs/2024/121/e3sconf_icrera2024_08005/e3sconf_icrera2024_08005.html

Evio, B. D., & Bonito, S. R. (2024). Formative evaluation of the implementation of eHealth in the Philippines: A qualitative study. *Acta Medica Philippina*, 58(12), 35–47. <https://doi.org/10.47895/amp.v58i12.9289>

Feldman, K. (2025, January 23). *A comprehensive guide to input-process-output models*. iSixSigma. Retrieved June 22, 2025, from <https://www.isixsigma.com/dictionary/input-process-output-i-p-o/>

Ghazaly, N. M., & Jain, N. (2023). IoT-based health monitoring system: Design, implementation, and performance evaluation. *Rajagiri Journal of Computer Science and Engineering*, 7(4), 1–12. <https://doi.org/10.52710/rjcse.84>

Guide, S. (2022, August 2). *What is a conceptual framework? | Tips & examples*. Scribbr. Retrieved June 22, 2025, from <https://www.scribbr.com/methodology/conceptual-framework/>

Jain, S. (2024, October 18). *Agile methodology advantages and disadvantages*. GeeksforGeeks. Retrieved June 22, 2025, from <https://www.geeksforgeeks.org/software-engineering/agile-methodology-advantages-and-disadvantages/>

GeeksforGeeks. (n.d.). *How to calculate the weighted mean*. Retrieved from GeeksforGeeks

Jonnalagadda, A., Rajvir, M., Singh, S., & Kamalov, F. (2022). An ensemble-based machine learning model for emotion and mental health detection. *Journal of Information & Knowledge Management*, 22(2), 2250075.

<https://doi.org/10.1142/S0219649222500757>

Joshi, A., Puricelli Perin, D. M., & Arora, M. (2013). Using portable health information kiosk to assess chronic disease burden in remote settings. *Rural and Remote Health*, 13(1), Article 2279. <https://doi.org/10.22605/RRH2279>

Kumar, S. (2024, January 24). *IoT based remote patient health monitoring system*. SciSpace. Retrieved June 24, 2026, from <https://scispace.com/papers/iot-based-remote-patient-health-monitoring-system-fxiqtwze46ig>

Kumar, S. R., Rao, T. C. S., Suman, J. V., & Rani, B. U. (2024). IoT based remote patient health monitoring system. *E3S Web of Conferences*, 591, Article 08005. <https://doi.org/10.1051/e3sconf/202459108005>

Lewis, J. R. (2018, March 30). *The system usability scale: Past, present, and future*.

ResearchGate. Retrieved June 24, 2025, from

https://www.researchgate.net/publication/324116412_The_System_Usability_Scale_Past_Present_and_Future

Maramba, I. D., Jones, R., Austin, D., Edwards, K., Meinert, E., & Chatterjee, A.

(2022). The role of health kiosks: A scoping review. *JMIR Medical*

Informatics, 10(3), e26511. <https://doi.org/10.2196/26511>

Marcelo, A. B., Silvestre, M. A. V., Palileo-Villanueva, L., Labarda, C., & Uy, H.

(2022). Factors affecting adoption of a telemedicine device by primary care health workers in the Philippines. *Acta Medica Philippina*, 56(12), 65–72.

<https://doi.org/10.47895/amp.vi0i0.3709>

Ng, W., Tipih, T., Makoah, N., Vermeulen, J., Goedhals, D., Sempa, J., Burt, F.,

Taylor, A., & Mahalingam, S. (2021). Comorbidities in SARS-CoV-2 patients:

A systematic review and meta-analysis. *mBio*, 12(1).

<https://doi.org/10.1128/mBio.03647-20>

Noceda, M. E., Ong, J. R., Nery, J. R., Racelis, S., & Mendoza, M. (2023). Patient

satisfaction with telemedicine in the Philippines during the COVID-19

pandemic: A mixed methods study. *BMC Health Services Research*, 23(1),

1331. <https://doi.org/10.1186/s12913-023-09127-x>

Profetto, L., Gherardelli, M., & Iadanza, E. (2022). Radio frequency identification

(RFID) in health care: Where are we? A scoping review. *Health and*

Technology, 12(5), 879–891. <https://doi.org/10.1007/s12553-022-00696-1>

Romanick-Schmiedl, S., & Raghu, G. (2020). Telemedicine—Maintaining quality

during times of transition. *Nature Reviews Disease Primers*, 6(1), 45.

<https://doi.org/10.1038/s41572-020-0185-x>

Saravanan, T. J., Mishra, M., Aherwar, A. D., & Lourenço, P. B. (2024). Internet of

things (IoT)-based structural health monitoring of laboratory-scale civil

engineering structures: Case study. *Innovative Infrastructure Solutions*, 9(1),

Article 110. <https://doi.org/10.1007/s41062-024-01413-9>

Saxena, A. (2023, September 14). *IoT-based patient wellness tracking system*.

SciSpace. Retrieved June 24, 2025, from [https://scispace.com/papers/iot-](https://scispace.com/papers/iot-based-patient-wellness-tracking-system-3pj1ql4087)

[based-patient-wellness-tracking-system-3pj1ql4087](https://scispace.com/papers/iot-based-patient-wellness-tracking-system-3pj1ql4087)

Saxena, A., Bhardwaj, A., Gupta, C., Kesarwani, R., & Bisariya, S. (2023). IoT-based

patient wellness tracking system. In *2023 International Conference on*

Sustainable Emerging Innovations in Engineering and Technology (ICSEIET)

(pp. 6–9). IEEE. <https://doi.org/10.1109/ICSEIET58612.2023.10156030>

SCRUMstudy. (2024, July 3). *Scrum methodology overview*. SCRUMstudy.com.

Retrieved June 24, 2025, from <https://www.scrumstudy.com/article/scrum-methodology-overview>

Sehgal, H. L. K., Greenfield, G., Neves, A. L., Harmon, M., Majeed, A., & Hayhoe, B. (2024). Efficacy and safety of a digital check-in and triage kiosk in emergency departments: A systematic review protocol. *BMJ Open*, 14(7), e084506. <https://doi.org/10.1136/bmjopen-2024-084506>

Sen, A., Dilektasli, A. G., Durak, V. A., Armagan, E., Korkmaz, E., Kiras, G., Baskan, E. B., & Temel, S. G. (2025). Use of a kiosk-model self-triage system for COVID-19 triage. *Risk Management and Healthcare Policy*, 18, 579–592. <https://doi.org/10.2147/RMHP.S497053>

Suleiman, T. A., & Adinoyi, A. (2023). Telemedicine and smart healthcare—The role of artificial intelligence, 5G, cloud services, and other enabling technologies. *International Journal of Communications, Network and System Sciences*, 16(3), 31–51. <https://doi.org/10.4236/ijcns.2023.163003>

Tolentino, L. K. S., Isoy, J. E. L., Bulawan, K. A. A., Co, M. C. T., Monreal, C. F. C., Vitto, I. J. W., Padilla, M. V. C., Quijano, J. F. C., Jorda, R. J. L., & Velasco, J. S. (n.d.). CAREdio: Health screening and heart disease prediction system for rural communities in the Philippines.

Villalvazo Laureano, E. (2024). An IoT health-care monitoring system design to measure vital parameters through non-invasive sensors. *Revista del Centro de Investigación de la Universidad La Salle*, 16(62), 51–72. <https://doi.org/10.26457/recein.v16i62.3644>

What is project planning? Definition and example. (2024, April 11). IdeaScale.

Retrieved June 30, 2025, from <https://ideascale.com/blog/what-is-project-planning/>

Xiang, J., Zhao, A., Tian, G. Y., Woo, W., Liu, L., & Li, H. (2022). Prospective RFID sensors for the IoT healthcare system. *Journal of Sensors*, 2022, Article 8787275. <https://doi.org/10.1155/2022/8787275>

Yadav, B. R. (2024). Smart RFID and IoT-based patient monitoring systems in modern healthcare. *International Journal of Engineering and Management Research*, 14(5), 89–93. <https://doi.org/10.5281/zenodo.14064940>



Appendices



APPENDIX A

Letters and Communication



Republic of the Philippines
Ilocos Sur Polytechnic State College- Candon Campus
Bachelor Of Science In Information Technology
Darapidap, Candon City, Ilocos Sur

August 13, 2025

MR. RICARDO BUGASTO
Barangay Captain
Brgy. Darapidap, Candon City, Ilocos Sur

Greetings!

We, the undersigned 4th year Bachelor of Science in Information Technology students of Ilocos Sur Polytechnic State College, Candon Campus, are currently enrolled in Capstone Project 2, conducting a study titled "CHECK: Community Health Evaluation Center Kiosk System".

Along with this, may we humbly request your good office to allow us gather relevant information about our study. Rest assured that all information gathered therein will be treated with utmost confidentiality and will be used for research purposes only.

We hope for your kind consideration and support. Thank you for your time and for the opportunity to work in collaboration with your esteemed office.

Respectfully yours,


VINCE MARC JUSTINE O. RUBANG


MELWYN MARC O. RAMOS


WINFREY CLINT O. LANGOEY


ASHLEY A. PASIBE

DIHON MARCK V. HERMOSURA

Student Researchers

Noted by:


MS. FE AIRAM R. DAKIG, MIT
Capstone Adviser

Recommending Approval:


MR. ARNEL A. DELA CRUZ, MIT
Program Head, BSIT

Approved by:

MR. RICARDO BUGASTO
Barangay Captain

AUG. 04, 2025

APPENDIX B

Evaluation/Tool/Instruments

To evaluate the system's acceptability, the researchers used an evaluation sheet based on ISO 9126-2. This tool measures six acceptability indicators: Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability. (ISO, 2003). Each indicator includes statements rated on a 7-point scale, from 1 (strongly disagree) to 7 (strongly agree). Respondents will include health workers, barangay officials, and residents.



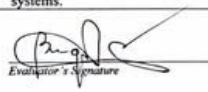

Republic of the Philippines
 Ilocos Sur Polytechnic State College- Candon Campus
 Bachelor of Science in Information Technology
 Darapidap, Candon City, Ilocos Sur

Evaluator's Profile
 Name: RICARDO M. BUGASTO
 Position: PUBLIC BARANGAY

Evaluator's Perception of the Developed Application
 Direction: Please evaluate the acceptability of the developed application "C.H.E.C.K (Community Health Evaluation Center Kiosk)" by using the given scale and placing a checkmark (✓) under the corresponding numerical rating:

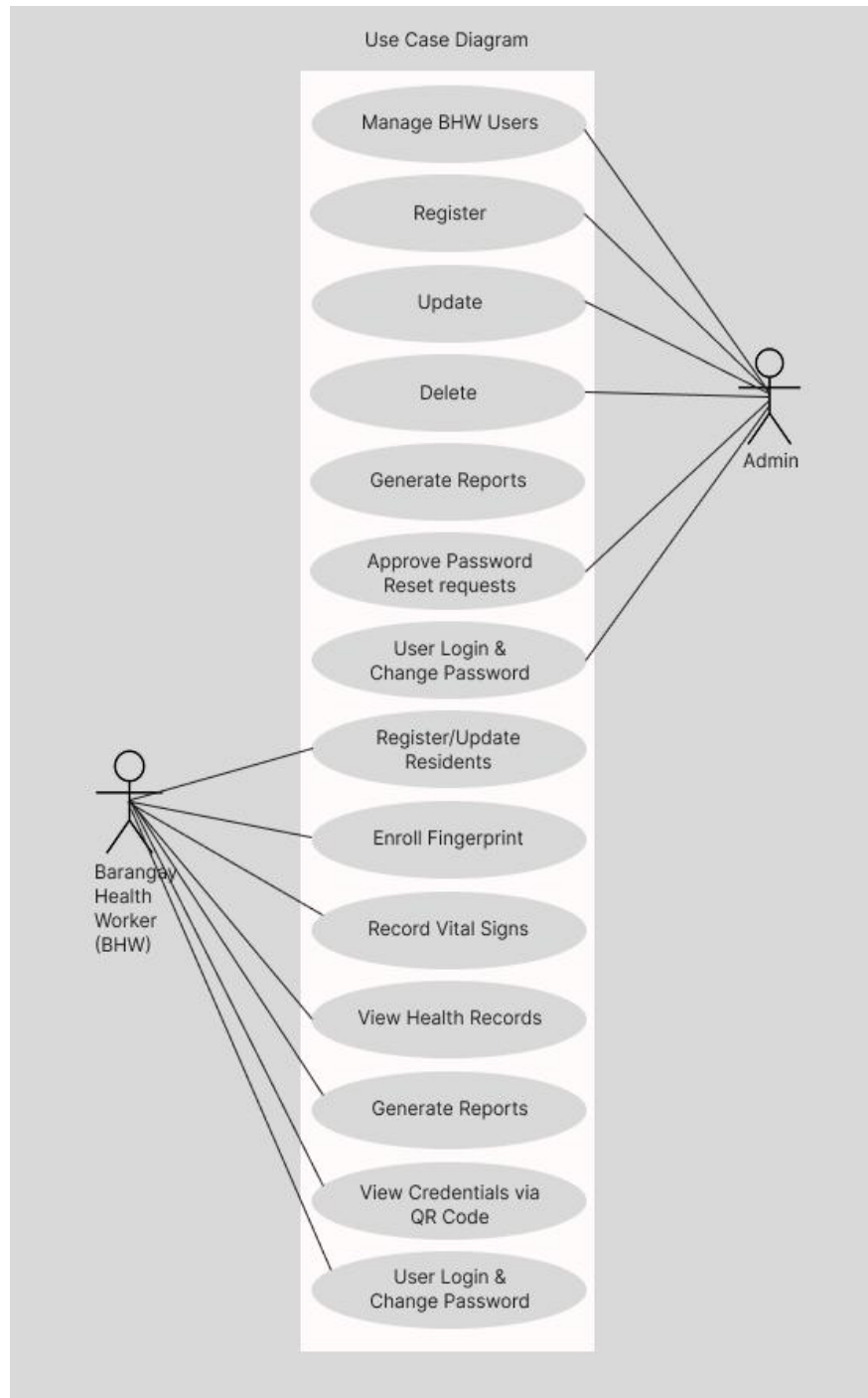
Numerical Rating: 7 – Strongly Agree
 6 – Agree
 5 – Slightly Agree
 4 – Neutral
 3 – Slightly Disagree
 2 – Disagree
 1 – Strongly Disagree

Acceptability Indicators	7	6	5	4	3	2	1
A. Functionality							
1. The system performs all intended functions correctly.		✓					
2. The system provides accurate results based on user input.		✓					
3. The system's features meet the user's needs.		✓					
B. Reliability							
1. The system works without unexpected crashes or errors.				✓			
2. The system gives consistent results during use.				✓			
3. The system can recover easily after errors or failures.				✓			
C. Usability							
1. The system is easy to use and understand.			✓				
2. The system's interface is user-friendly.			✓				
3. The system helps users' complete tasks quickly and correctly.			✓				
D. Efficiency							
1. The system loads and responds quickly.			✓				
2. The system performs tasks without unnecessary delays.			✓				
E. Maintainability							
1. The system is easy to update or fix when needed.			✓				
2. The system's code or structure is well organized.			✓				
3. The system allows adding new features easily.			✓				
F. Portability							
1. The system can run on different devices or platforms.			✓				
2. The system works properly after being moved to another environment.			✓				
3. The system needs only small changes to work on other systems.			✓				


 Evaluator's Signature

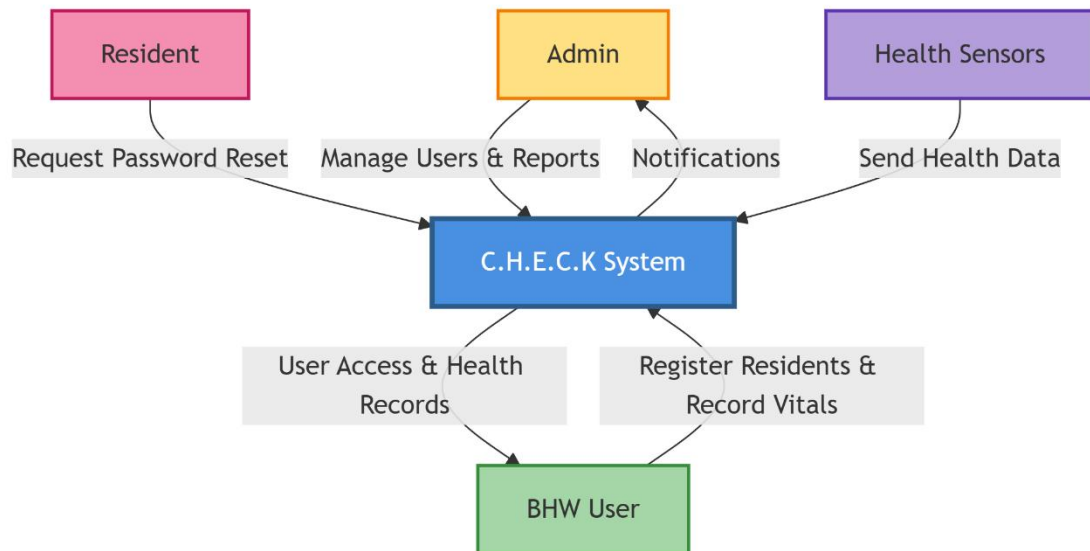
APPENDIX C

Use Case Diagram



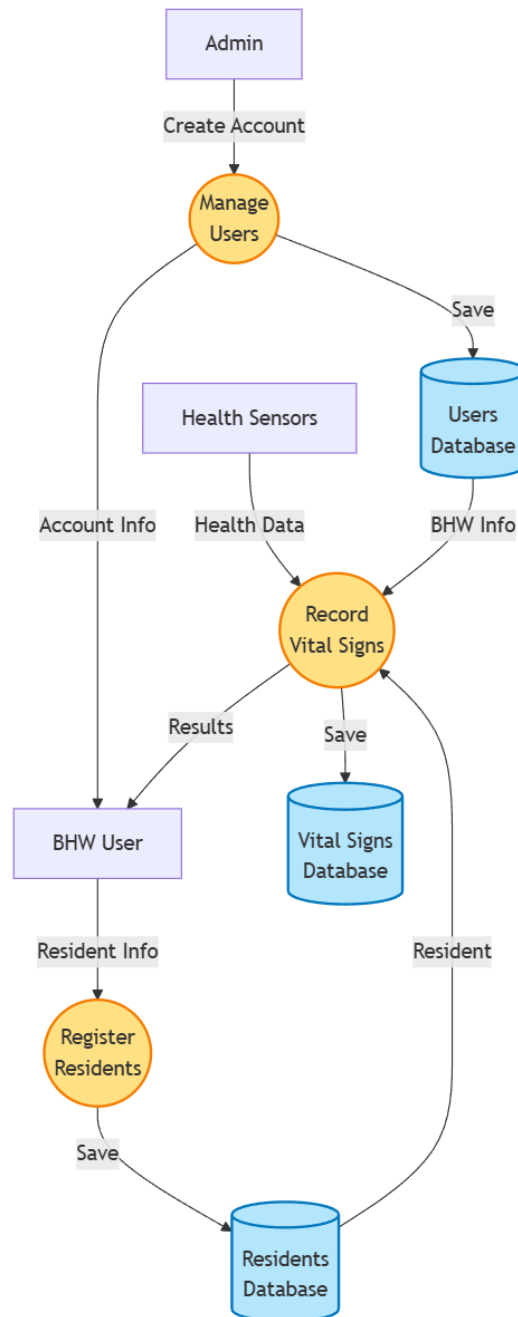
APPENDIX D

Context Diagram



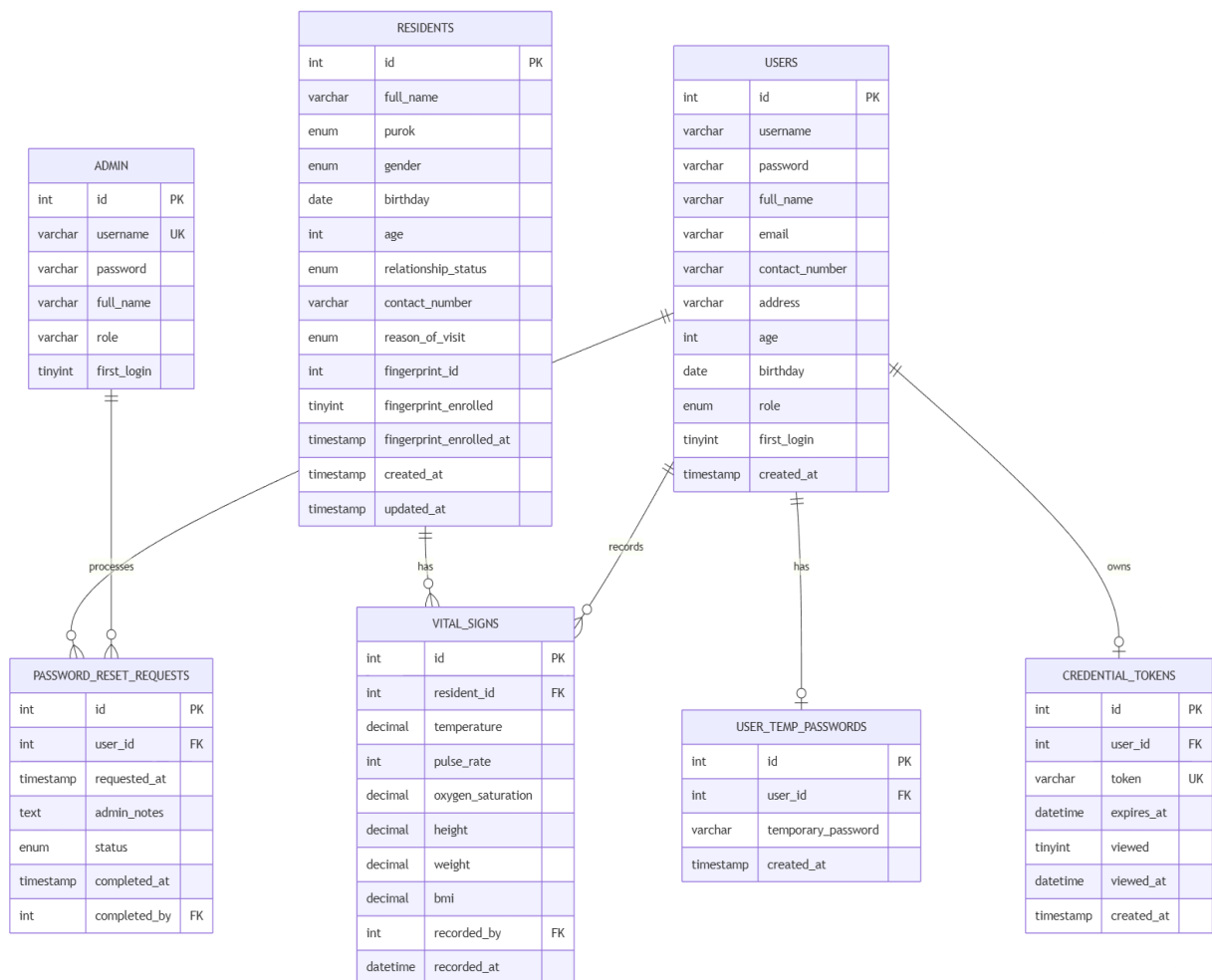
APPENDIX E

Data Flow Diagram



APPENDIX F

Entity Relationship Diagram



APPENDIX G

User Manual

Community Health Evaluation Center Kiosk (C.H.E.C.K) User Manual provides an overview of how to use the C.H.E.C.K health monitoring system effectively. It covers essential tasks such as resident management, BHW user management, fingerprint enrollment, and vital signs monitoring. The manual also explains how to manage user profiles, access key features like biometric identification, and navigate system functionalities. The goal is to help users fully leverage C.H.E.C.K's capabilities to enhance the efficiency of health data management in Barangay Darapidap

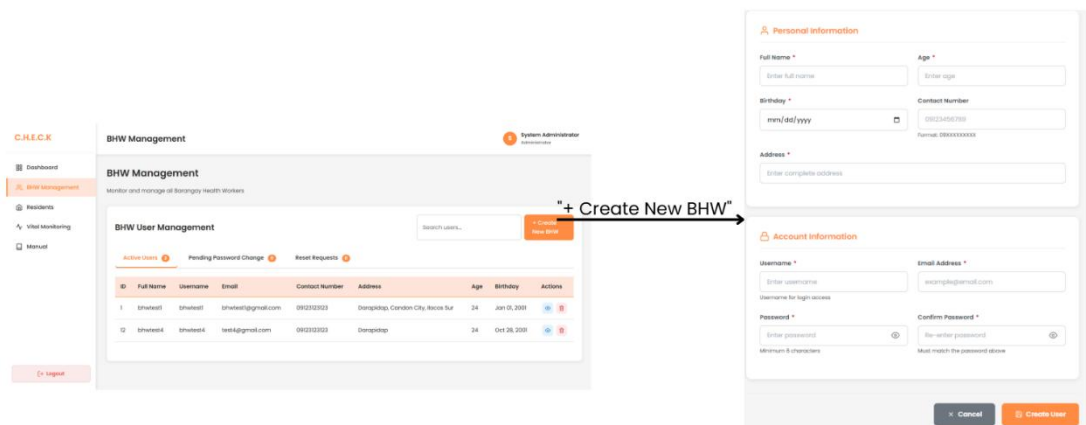
Login - To access the system, users must enter their unique username and password on the login page. The system displays the C.H.E.C.K logo with "Community Health Evaluation Center Kiosk" as the header. Upon successful login, the System Administrator or User will be directed to their dashboard, which provides a comprehensive overview of key system statistics.



Forgot Password - If the user forgets their login credentials, they can click the "Forgot Password? Request Reset" link located below the password field. This will redirect them to the Request Password Reset page where they must input their username. The system will display a "Password Reset Process" guide explaining that the request will be submitted to the system administrator, who will review the request in BHW Management. Each reset request entry shows a Status indicator, typically displaying "Pending" in an orange badge to signify that the request awaits administrative action.

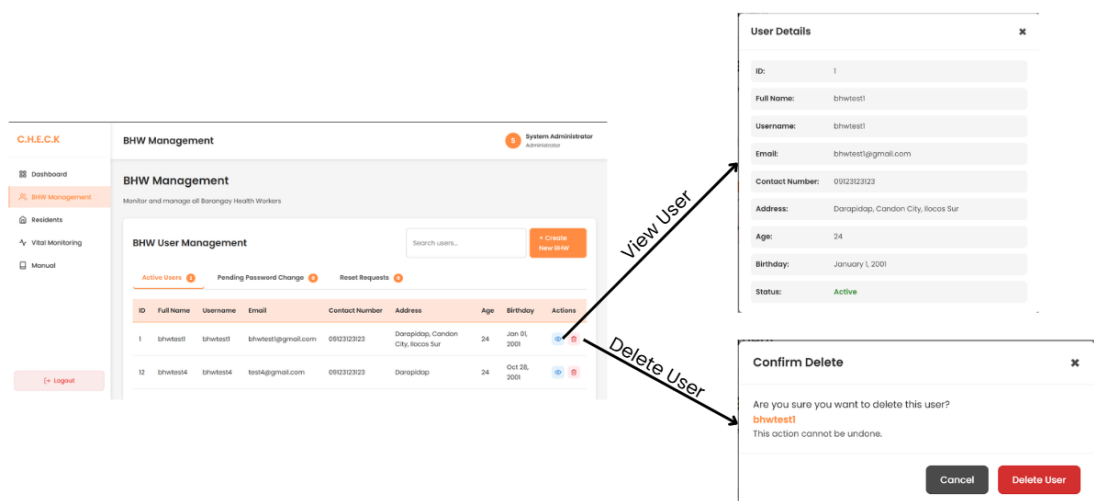


Create New BHW User - To create a new BHW user, click the "+ Create New BHW" button located in the upper right corner of the BHW Management page. A registration form will appear prompting the administrator to fill out all required fields.



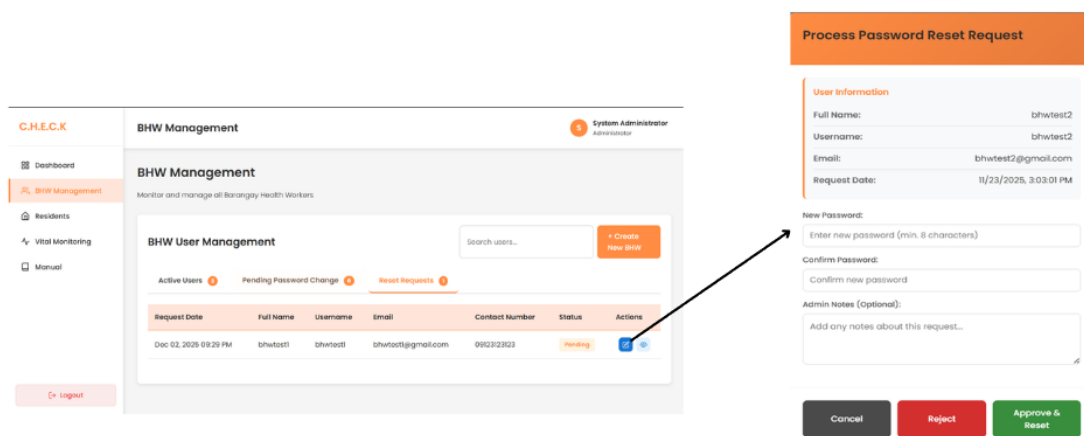
The screenshot shows the BHW Management interface. On the left is a sidebar with a 'CHECK' section and a 'BHW Management' section. The main area displays the 'BHW User Management' table with columns: ID, Full Name, Username, Email, Contact Number, Address, Age, Birthday, and Actions. An orange '+ Create New BHW' button is in the top right of the table area. An arrow points from this button to a registration form on the right. The form has two sections: 'Personal Information' and 'Account Information'. The 'Personal Information' section includes fields for Full Name, Age, Birthday, Contact Number, and Address. The 'Account Information' section includes fields for Username, Email Address, Password, and Confirm Password. At the bottom right of the form are 'Cancel' and 'Create User' buttons.

Viewing and Deleting BHW Users - Each BHW entry in the Active Users table has action icons displayed on the right side. The View Icon (eye symbol) allows administrators to see complete BHW information including login status and verification details. The Delete Icon (trash bin symbol) enables permanent removal of the BHW user from the system.



The screenshot shows the BHW Management interface with the 'BHW User Management' table. An arrow labeled 'View User' points from the 'View' icon (eye symbol) in the Actions column of the first row to a 'User Details' modal. Another arrow labeled 'Delete User' points from the 'Delete' icon (trash bin symbol) in the Actions column of the second row to a 'Confirm Delete' modal. The 'User Details' modal displays the following information: ID: 1, Full Name: bhwtest1, Username: bhwtest1, Email: bhwtest1@gmail.com, Contact Number: 0912312323, Address: Darapidap, Candon City, Ilocos Sur, Age: 24, Birthday: January 1, 2001, and Status: Active. The 'Confirm Delete' modal asks: 'Are you sure you want to delete this user? bhwtest1. This action cannot be undone.' and has 'Cancel' and 'Delete User' buttons.

Processing a Password Reset Request - To process a password reset request, click the Process Request icon in the Actions column. The system displays the "Process Password Reset Request" screen with an orange header. The User Information section displays the Full Name, Username, Email, and Request Date with timestamp showing exactly when the BHW submitted the password reset request. This information helps administrators verify the identity of the requesting user. The New Password field requires administrators to enter a new temporary password with a minimum of eight characters. The system provides a placeholder text stating "Enter new password (min. 8 characters)." to guide proper password creation.



Process Password Reset Request

User Information

Full Name: bhwtest2
Username: bhwtest2
Email: bhwtest2@gmail.com
Request Date: 11/23/2025, 3:03:01 PM

New Password:
Enter new password (min. 8 characters)

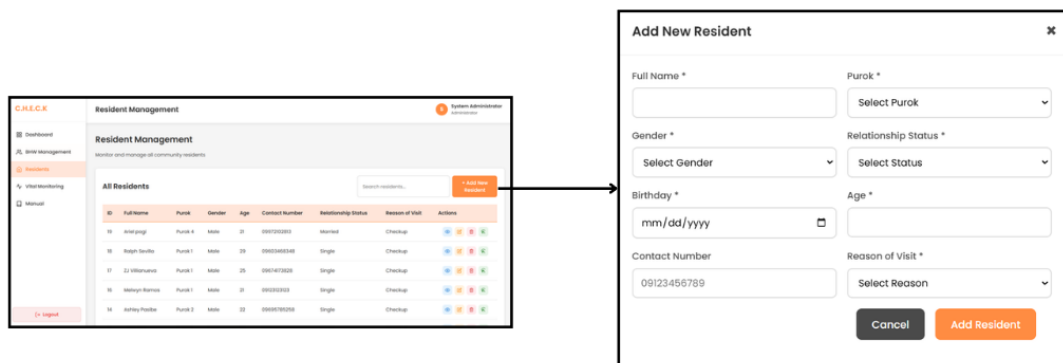
Confirm Password:
Confirm new password

Admin Notes (Optional):
Add any notes about this request...

Buttons: Cancel, Reject, Approve & Reset

Add New Resident - To register a new resident, click the "+ Add New Resident" button in the upper right corner of the Resident Management page. The system will display an "Add New Resident" form modal containing several required fields.

Resident Details - Each resident entry in the table features an eye icon in the



The image shows a screenshot of the 'Resident Management' interface. On the left, a sidebar contains navigation links: Dashboard, Resident Management, Residents, Visit Scheduling, and Manual. The main area displays a table titled 'All Residents' with columns: ID, Full Name, Purok, Gender, Age, Contact Number, Relationship Status, Reason of Visit, and Actions. An arrow points from the 'Add New Resident' button in the top right of the table to a modal form titled 'Add New Resident'. The modal contains the following fields: Full Name (text input), Purok (dropdown menu), Gender (dropdown menu), Relationship Status (dropdown menu), Birthday (date picker), Age (text input), Contact Number (text input), and Reason of Visit (dropdown menu). At the bottom of the modal are 'Cancel' and 'Add Resident' buttons.

ID	Full Name	Purok	Gender	Age	Contact Number	Relationship Status	Reason of Visit	Actions
10	Arnel Angil	Purok 4	Male	21	0907300203	Married	Checkup	  
18	Adrian Sallida	Purok 11	Male	39	0903040304	Single	Checkup	  
21	Jo Villanueva	Purok 11	Male	25	0901473029	Single	Checkup	  
19	Melvin Barrios	Purok 11	Male	21	0903030303	Single	Checkup	  
14	Anthony Probito	Purok 2	Male	22	0906790208	Single	Checkup	  

Actions column. Clicking this View Details icon opens a modal displaying complete resident information including Full Name, Purok, Gender, Birthday, Age, Relationship Status, Contact Number, Reason of Visit, Fingerprint Status, and Registered Date. The Fingerprint Status indicates whether the resident has been enrolled in the biometric system, showing either "Enrolled" with an ID number or "Not Enrolled."

Edit Resident Info - To update resident information, click the Edit icon (pencil symbol) in the Actions column. The "Edit Resident" form modal appears, pre-populated with the current resident data. Administrators can modify any field including Full Name, Purok, Gender, Relationship Status, Birthday, Age, Contact Number, and Reason of Visit. After making the necessary changes, click the "Update Resident" button to save the modifications. Click "Cancel" to close the form without applying any changes.

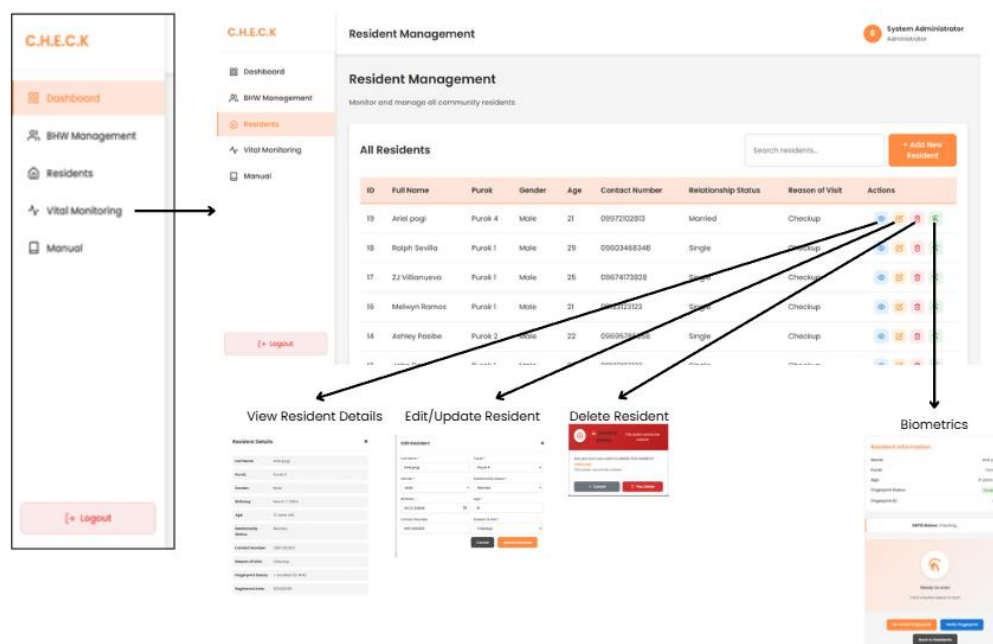
Delete Resident Records - To remove a resident from the system, click the Delete icon (trash bin symbol) in the Actions column. A confirmation dialog appears titled "Confirm Delete" asking "Are you sure you want to delete this resident?" with the resident's name displayed in orange text. The dialog warns that "This action cannot be undone." Click the "Delete Resident" button to permanently remove the resident record, or click "Cancel" to abort the deletion process and return to the Resident Management page.

Fingerprint Enrollment - The Fingerprint Enrollment feature enables biometric identification of residents using fingerprint sensors. This feature integrates with an ESP32 device to capture and store fingerprint data securely. From the Resident Management page, locate the resident who needs fingerprint enrollment. Click the green Fingerprint icon in the Actions column. The system will redirect to the Fingerprint Enrollment page.

The Fingerprint Enrollment page displays Resident Information at the top, showing the Name, Purok, Age, and current Fingerprint Status. If the resident has not been enrolled, the status displays "Not Enrolled" in an orange badge. The interface shows the ESP32 Status indicator which displays the current connection state of the fingerprint sensor device. The status updates to show "Checking..." while verifying the connection, and "Connected" when the device is ready to scan. Below the status indicator is a large fingerprint icon with the text "Ready to scan" and instructions stating "Click a button below to start." This visual interface helps guide users through the enrollment process.

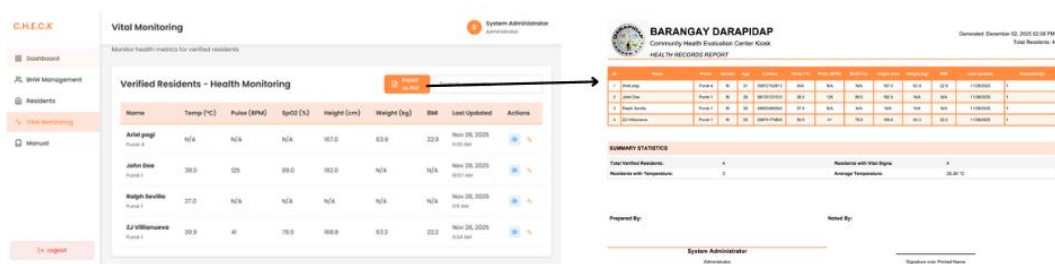
To begin the enrollment process, click the "Enroll Fingerprint" button. The system will communicate with the ESP32 device and prompt the resident to place their finger on the sensor. The interface will guide through a two-step scanning process.

Vital Monitoring - The Vital Monitoring section tracks and records health metrics for residents who have completed fingerprint enrollment. This feature requires integration with health monitoring devices to capture vital signs data accurately and efficiently.



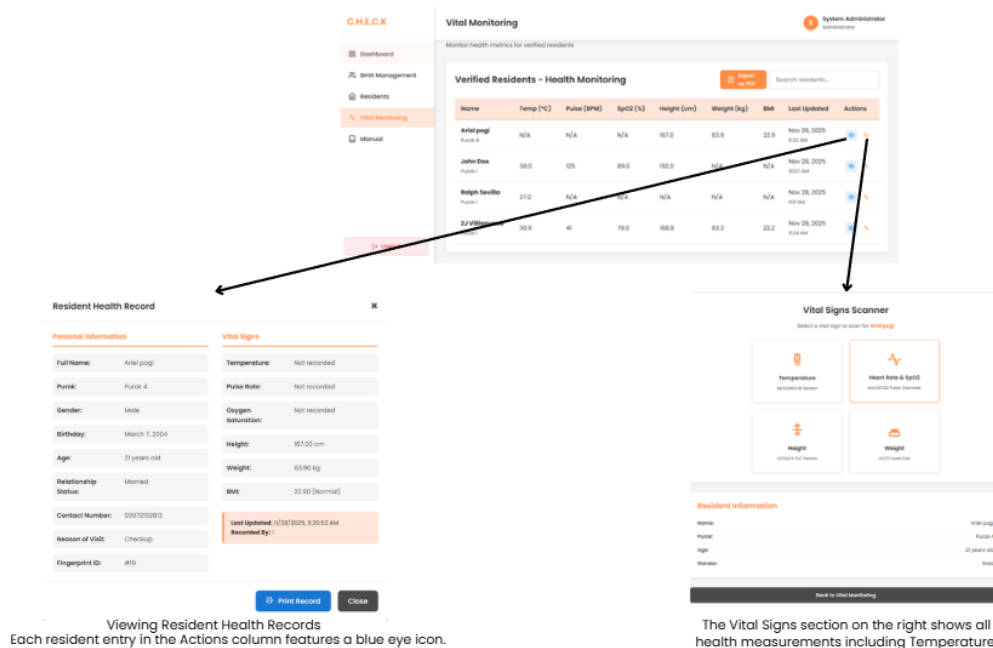
Viewing Resident Health Records - Each resident entry in the Actions column features a blue eye icon. Clicking this View icon opens the "Resident Health Record" modal displaying complete information about the selected resident. The modal is divided into two main sections. The Personal Information section on the left displays Full Name, Purok location, Gender, Birthday with full date, Age in years, Relationship Status (Single, Married, etc.), Contact Number in standard format, Reason of Visit, and Fingerprint ID showing the assigned biometric identification number.

The Vital Signs section on the right shows all health measurements including Temperature, Pulse Rate, Oxygen Saturation, Height, Weight, and BMI. For residents without recorded vital signs, each field displays "Not recorded" or "Not calculated" for BMI. At the bottom of the modal, two action buttons are available. The blue "Print Record" button generates a printable version of the health record. The gray "Close" button closes the modal and returns to the Vital Monitoring page.



The screenshot shows the 'Vital Monitoring' interface. On the left is a sidebar with 'CHECK' and 'Vital Monitoring' sections. The main area displays a table titled 'Verified Residents - Health Monitoring' with columns: Name, Temp (°C), Pulse (BPM), SpO2 (%), Height (cm), Weight (kg), BMI, Last Updated, and Actions. The table lists four residents: Arnel pongi, John Doe, Ralph Saville, and Zz Villanueva. An orange 'Export as PDF' button is located at the top right of the table. To the right of the table is a 'BARANGAY DARAPIDAP' summary report showing statistics for verified residents and those with temperature readings.

Exporting Data as PDF - In the upper right corner of the Verified Residents table, an orange "Export as PDF" button allows users to generate comprehensive

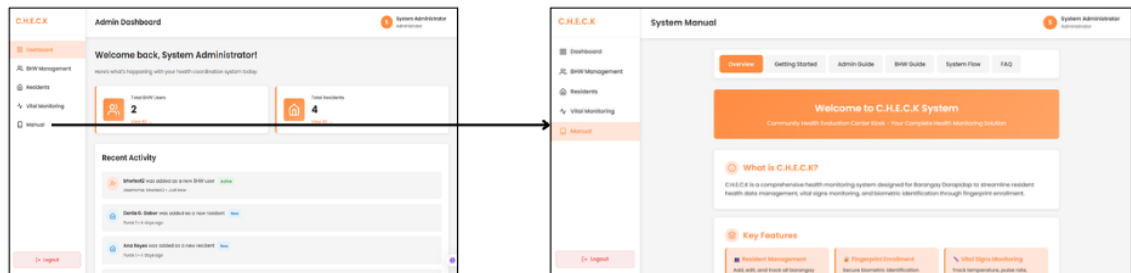


The screenshot shows the 'Resident Health Record' modal. It is divided into two sections: 'Personal Information' and 'Vital Signs'. The 'Personal Information' section includes fields for Full Name, Purok, Gender, Birthday, Age, Relationship Status, Contact Number, Reason of Visit, and Fingerprint ID. The 'Vital Signs' section includes fields for Temperature, Pulse Rate, Oxygen Saturation, Height, Weight, and BMI. Each field in the 'Vital Signs' section has a status indicator: 'Not recorded' or 'Not calculated'. At the bottom of the modal are two buttons: 'Print Record' and 'Close'. To the right of the modal is a 'Vital Signs Scanner' section with a search bar and four buttons: 'Temperature', 'Heart Rate & SpO2', 'Height', and 'Weight'.

Viewing Resident Health Records
Each resident entry in the Actions column features a blue eye icon.

The Vital Signs section on the right shows all health measurements including Temperature, Pulse Rate, Oxygen Saturation, Height, Weight, and BMI

reports. Clicking this button exports all verified residents' health monitoring data into a downloadable PDF document for printing, archival, or sharing purposes.



System Manual - The System Manual provides comprehensive documentation and guides for using the C.H.E.C.K system. Access this section by clicking "Manual" from the sidebar.

APPENDIX H

Actual and Final Design of C.H.E.C.K Kiosk





APPENDIX I

Certificate of English Critic

ILOCOS SUR POLYTECHNIC STATE COLLEGE
CANDON CAMPUS
City of Candon, Ilocos Sur

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

CERTIFICATION

This is to certify that I have checked the capstone project manuscript copy of:

Vince Marc Justine O. Rubang, Winfrey Clint O. Langoey, Dhonmarck Hermosura, Ashley
proponents name

A. Pasibe, Melwyn Marc O. Ramos

entitled "C.H.E.C.K (Community Health Evaluation Center Kiosk)", which has been
capstone title
duly noted as requirements for graduation in BSIT Program.

This certification is issued this 17th day of December, 2025 at ILOCOS
date month year
SUR POLYTECHNIC STATE COLLEGE CANDON CAMPUS, ILOCOS SUR.


Mr. Ralph Nicole Sevilla
English Critic



APPENDIX J

Certificate of Statistician

ILOCOS SUR POLYTECHNIC STATE COLLEGE
CANDON CAMPUS
City of Candon, Ilocos Sur

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY CERTIFICATION

This is to certify that I have checked the capstone project manuscript copy of:

Vince Marc Justine O. Rubang, Winfrey Clint O. Langoey, Dhonmarck Hermosura,
proponents name

Ashley A. Pasibe, Melwyn Marc O. Ramos

entitled "C.H.E.C.K (Community Health Evaluation Center Kiosk)", which has been
capstone title
duly noted as requirements for graduation in BSIT Program.

This certification is issued this 17th day of December, 2025 at ILOCOS
date month year
SUR POLYTECHNIC STATE COLLEGE CANDON CAMPUS, ILOCOS SUR.


Mr. ZG G. Villanueva
Statistician



APPENDIX K

Certificate of Implementation



Republic of the Philippines
Ilocos Sur Polytechnic State College- Candon Campus
Bachelor Of Science in Information Technology
Darapidap, Candon City, Ilocos Sur

CERTIFICATION

This is to certify that the system titled **"C.H.E.C.K: COMMUNITY HEALTH EVALUATION CENTER KIOSK"** created by **VINCE MARC JUSTINE O. RUBANG, WINFREY CLINT O. LANGOEY, DHON MARCK V. HERMOSURA, MELWUN MARC O. RAMOS, AHSLEY A. PASIBE** has been successfully implemented as a requirement for graduation in the BSIT program.

This certification is issued this month of **DECEMBER, 2025** at the **ILOCOS SUR POLYTECHNIC STATE COLLEGE CANDON CAMPUS, CITY OF CANDON, ILOCOS SUR.**


MR. RICARDO BUGASTO
Barangay Captain

APPENDIX L

PICTURES TAKEN DURING DATA GATHERING, PLANNING AND DEVELOPMENT



Data Gathering Phase



Planning and Development Phase

BIOGRAPHICAL SKETCH

Vince Marc Justine O. Rubang, born on October 28, 2001 in Candon City, Ilocos Sur, is a dedicated and driven young professional known for his adaptability, flexibility, and strong willingness to learn. Raised in San Jose, Candon City, he is the middle child of Evelyn O. Rubang and Melvin G. Rubang, whose support shaped his commitment to excellence and personal growth.



Vince has developed a deep passion for technology and system development, pursuing both software and hardware with equal focus. In his projects, he has taken on the crucial roles of main developer, project manager, and primary writer, a testament to his leadership, initiative, and comprehensive skill set. He proudly acknowledges his contributions, having carried the bulk of the development work with diligence and vision.

Known for being coachable, teachable, and consistently eager to grow, Vince thrives in environments that challenge his creativity and technical abilities. His adaptive mindset allows him to approach complex tasks with confidence, while his flexibility enables him to collaborate effectively across different aspects of system development. At 24, Vince continues to build a strong foundation in his field, guided by curiosity, discipline, and an unwavering commitment to producing meaningful and high-quality work.



BIOGRAPHICAL SKETCH

Winfrey Clint Langoey was born on November 28, 2002, in Apdo Lintic, Banayoyo, Ilocos Sur. He is the son of Edwin Langoey and Ela Langoey. Growing up in a simple community, Winfrey learned the value of hard work, perseverance, and staying true to his goals.



He completed his primary education at Banayoyo Central School and continued his junior and senior high school studies at Banayoyo National High School. At present, he is a fourth-year college student pursuing a Bachelor of Science in Information Technology. Throughout his academic journey, Winfrey gradually developed an interest in technology, particularly in areas such as prototyping, web design, and web development.

Outside of academics, Winfrey enjoys playing basketball during his free time, which helps him relax and stay active. His skills in IT continue to grow as he engages in projects that challenge his creativity and technical abilities. Looking toward the future, Winfrey hopes to build his own tech company one day. His ambition is driven by his desire to create innovative solutions and contribute to the growth of the technology industry. With passion, dedication, and continuous learning, Winfrey believes he can achieve his goals and make a meaningful impact.

BIOGRAPHICAL SKETCH

Ashley A. Pasibe was born on October 26, 2003. Pasibe spent his early years in Sitio Nazareth, Valley 6, Parañaque City before his family relocated to Poblacion Este, Sta. Cruz, Ilocos Sur in 2017. He completed his primary education at CAA Elementary School and his secondary education at Dili National High School, where his team achieved a victory in a basketball inter-high competition. For his senior high school studies, he pursued the Information and Communications Technology (ICT) strand and went on to proudly represent his school at the regional level in the sport of sepak takraw.



Pasibe is currently a fourth-year student at Ilocos Sur Polytechnic State College – Candon Campus, pursuing a Bachelor of Science in Information Technology. His interest in ICT developed as he recognized the continuous innovations shaping the digital world and the opportunities the field offers. Although he initially faced challenges, he learned to appreciate the process, which subsequently deepened his passion for technology. He is known for his strong work ethic, discipline, and dedication. His experiences in both academics and sports have significantly strengthened his determination and perseverance to excel. His commitment to lifelong learning and personal growth continues to guide him as he works toward building a successful future in the field of Information Technology.

BIOGRAPHICAL SKETCH

Mr. Melwyn Marc O. Ramos is a 21-year-old student at Ilocos Sur Polytechnic State College – Candon Campus, pursuing a Bachelor of Science in Information Technology. He was born in Pangasinan but was raised in Ilocos Sur, where he developed an early interest in computers. This initial curiosity later



inspired him to pursue a degree in IT, with a focus on programming, hardware integration, and system development.

Throughout his academic journey, Mr. Ramos has actively participated in various school activities and group projects, which have strengthened his skills in web development and IoT systems. He enjoys working on practical, community-focused projects, particularly those that utilize technology to solve real-life problems. A significant example of this is his contribution to the development of the C.H.E.C.K: IoT Health Check Kiosk, where he primarily focused on system testing, documentation, and user evaluation. He is committed to continuously learning and improving his technical abilities. In the future, Mr. Ramos aspires to become a professional developer or systems analyst capable of creating innovative solutions that benefit both local communities and the broader society.

BIOGRAPHICAL SKETCH

Dhon Marck V. Hermosura was born on March 10, 2004. He is the son of Alice Joy V. Hermosura and Ricardo Hermosura. Growing up under the guidance of his loving parents, Dhon Marck learned the values of responsibility, perseverance, and determination, which continue to shape his character as a young individual.



He completed his primary education at Saint Joseph Sahagun in Candon City. He then finished his elementary education at Santa Lucia South Central School. For his secondary education, he completed both his high school and senior high school at Santa Lucia Catholic School, where he strengthened his academic foundation and developed discipline and dedication toward his studies.

At present, he is a fourth-year college student pursuing the degree of Bachelor of Science in Information Technology. Throughout his academic journey, he has developed a strong interest in technology, particularly in gaming, video editing, and photo editing, which allow him to express his creativity while improving his technical abilities. Through his educational journey and personal interests, Dhon Marck continues to strive for personal growth and success. With the values he learned from his family and school, he remains focused on achieving his goals and building a better future for himself.